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Martian Ages

G. NEUKUM AND K. HILLER

Institut für Allgemeine und Angewandte Geologie, Ludwig-Maximilians-Universität München
8000 München 2, Federal Republic of Germany

The subjects of this paper are a discussion of the methodology of relative age determination by impact crater statistics, a comparison of currently proposed Martian impact chronologies for the determination of absolute ages from crater frequencies, a report on our work of dating Martian volcanoes and erosional features by impact crater statistics, and an attempt to understand the main features of Martian history through a synthesis of our crater frequency data and those published by other authors. Two cratering chronology models are presented and used for inference of absolute ages from crater frequency data: model I, with nearly equal Martian and lunar cratering rates around (ca.) 4- to 10-km crater sizes, and model II, equivalent to model I for ages $> 3.5 \cdot 10^9$ years but with a factor of 2 higher Martian cratering rate at ages $< 3 \cdot 10^9$ years. Those model cratering chronologies are applied to the data. The interpretation of all crater frequency data available and tractable by our methodology leads to a global Martian geological history that is characterized essentially by two epochs of activity. The division between the two epochs is measured at a cumulative crater frequency value for 1-km craters (crater retention age) of $N(1) = 8 \cdot 10^{-4}$ (km^{-2}) corresponding to an absolute age of ca. $3 \cdot 10^9$ years (applying model I cratering chronology) and of ca. $1.5 \cdot 10^9$ years (applying model II cratering chronology). In the ancient epoch all major events like emplacement of the plains lavas, the piling up of most volcanic constructs, and large-scale erosion of channels and mensae (highland/northern lowland boundary) have taken place. During the younger epoch, only the big Tharsis shield volcanoes were active, and some minor erosion took place. This means that Mars is not a youthful planet but an ancient one with respect to most of its surface features.

INTRODUCTION

Problems of importance in the investigation of Mars are the time of occurrence of geologic processes like volcanism and large-scale erosion, the sequence of events, and correlation of processes. Impact crater statistics provide a tool for establishing relative and, within model-dependent limitations, absolute time scales.

Very different Martian absolute cratering chronologies have been proposed in the literature, as reviewed by *Neukum and Wise* [1976*a*]. The application of these different model chronologies to relative ages as inferred from crater counts resulted in substantially different histories of the Martian crust. Recently, the reevaluation of the problem of the ratio of Martian to lunar impact rates and of the problem of the relationship of the lunar cratering chronology to the Martian one has led to a revision in some models [*Soderblom*, 1977; *Hartmann*, 1978]. The magnitude of the changes and the remaining differences in the models will be discussed as well as the reasons for the discrepancies in the models.

This report presents the results of our efforts to determine crater size-frequency distributions for a variety of areas, to intercompare crater statistics data of different authors, and to extract relative and absolute ages from crater frequencies. The analysis of great data numbers obtained from different regions will lead to a better understanding of the sequence and interaction of different geologic events and eventually of Martian global history. It has to be understood that the data used in the analysis here are partially of very different origin, i.e., from a great number of authors using very different methods with respect to target selection, data acquisition and evaluation, accuracy, etc. We have treated all data in the same way, our own and those of others, by application of our crater statistics methodology. For the data extracted from the literature this has necessarily meant a reevaluation. Only in this way have we been able to arrive at a set of coherent and roughly comparable data.

THE MARTIAN IMPACT CRATER PRODUCTION SIZE-FREQUENCY DISTRIBUTION

The basic method of determining relative ages of different areas by crater statistics is to compare frequencies of impact craters superimposed on these areas. A premise in this procedure is that one deals with the undisturbed image (i.e., the craters) of the mass and velocity distribution of the impactors. A common problem on planetary surfaces such as that of Mars, however, is the interaction of geologic processes with cratering, resulting in size-selective destruction of craters. Therefore the distributions one finds on Mars typically do not show the undisturbed production size-frequency state but have recorded resurfacing effects. Those effects show up in irregularities in the distributions ('bumps') or flattening toward smaller sizes. Such irregularities mean that different sizes of craters had different retention times and therefore one measures different retention ages at different sizes, i.e., different ratios of actually determined crater frequencies to the production size-frequency distribution at different diameters.

Irregularities in the distributions can also be produced by different target response to cratering due to factors such as different lithologies, contrasting strength of the target substrate for craters of different sizes, water content, layering, etc., as discussed for the lunar and/or Martian case by *Young et al.* [1974], *Schultz and Spencer* [1979], *Boyce and Roddy* [1978], *Boyce* [1979], *Mouginis-Mark* [1979], or *Croft et al.* [1979]. *Neukum and Wise* [1976*a*, *b*] also realized the possibility of such target effects on crater sizes and discussed the effects on crater size-frequency distributions but considered them small for the Martian case but possibly present. However, as demonstrated in some of the recent work cited, the water content in the Martian target material could well be responsible for producing some of the irregularities in the distributions at cra-

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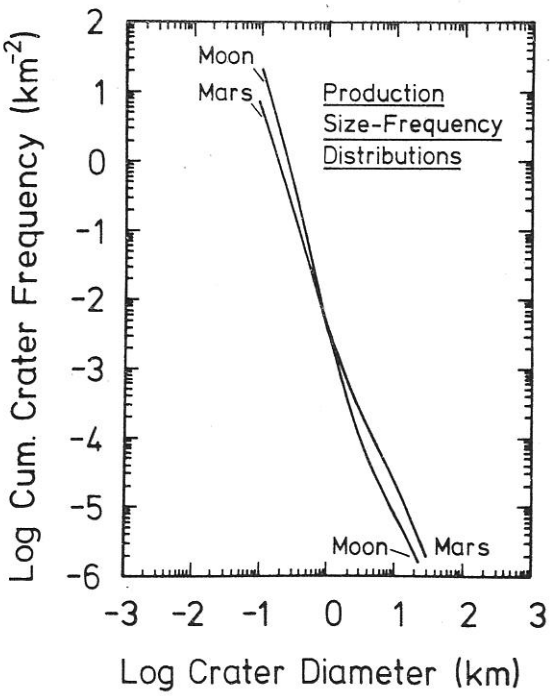


Fig. 1. Lunar and Martian cumulative impact crater production size-frequency distributions (standard curves, reference distributions).

ter sizes larger than 2–6 km. It is very difficult to disentangle the resurfacing and target response effects in the Martian distributions especially if they both happened to various extents, as may have been the case in Martian cratering history. We still consider resurfacing to be largely responsible for the irregularities found in the Martian crater size-frequency distributions as discussed later (Figures 5–8). If this is a misinterpretation, and the difference in target response is the major reason for the irregularities at crater sizes larger than 2–6 km, then we would have made an error of about a factor of 2 in the determination of some of the respective crater frequencies [cf. *Boyce,* 1979]. Another basic problem is that often it is not possible to compare frequencies of craters of different areas at the same crater sizes because of statistical limitations (e.g., low frequency of large craters in young areas or in small counting areas) or because of erosional problems with small craters or contamination of small crater populations by secondary craters, etc.

The derivation of meaningful relative ages ('crater retention ages' [*Hartmann,* 1966]) from a set of measured data therefore demands in many cases the comparison with a production size-frequency distribution. In the case of Mars, this distribution was first determined by *Neukum and Wise* [1976*a*] for the size range 0.8 km $\leq D \leq$ 20 km. The distribution hereafter termed 'standard crater size-frequency distribution' or 'standard distribution' or 'standard crater curve' has been extended to crater sizes smaller than 1 km [*Neukum et al.,* 1978]. It has subsequently been tested against a variety of measured Martian distributions especially at sizes $D \leq 1$ km. The distribution has been found to be somewhat flatter at sizes $D \leq 1$ km than maintained by *Neukum et al.* [1978]. A detailed discussion will be given elsewhere. The result is shown in Figure 1 in comparison with the lunar curve.

The Martian standard distribution is believed to represent the time-invariant Martian production size-frequency distribution, considered valid within a factor of 2 uncertainty in frequency through the size range 0.1 km $\leq D \leq$ 20 km. It can be approximated analytically by a polynomial of the form

$$\log N = a_0 + a_1 \log D + a_2(\log D)^2 + \cdots + a_7(\log D)^7$$

where D means crater diameter in km and N means cumulative crater frequency per km^2, a_0 is variable and gives the time dependence (i.e., the relative age), which means a vertical (N direction) shift of the curve in a log N versus log D diagram such as in Figure 1 or 2 according to the difference in relative ages (no shift in D direction!). The coefficients a_1 to a_7 are

$$\begin{aligned} a_1 &= -2.966 & a_5 &= -0.193 \\ a_2 &= \ \ 0.792 & a_6 &= \ \ 0.250 \\ a_3 &= \ \ 0.485 & a_7 &= -0.050 \\ a_4 &= -0.577 \end{aligned}$$

The polynomial given here is largely identical in the diameter range 1 km $\leq D \leq$ 20 km with that published by *Neukum and Wise* [1976*a*].

As seen in Figure 1, the Mars standard distribution has an overall flatter appearance than the lunar one. Therefore comparison of lunar crater frequencies with martian ones cannot be done independently of crater size. Possible reasons for the difference in the shapes of the lunar and Martian standard distributions have been discussed by *Neukum and Wise* [1976*a*], the most probable being a difference in impact velocity (about a factor of 1.5 lower on Mars), which would produce just the observed differences in the diameter range 1 km $\leq D \leq$ 20 km. In the diameter range 0.1 km $< D <$ 1 km, however, a difference of a factor of 1.5 in impact velocity would not produce the observed difference in the distributions, but rather a factor of ~2. Currently, our data are not good enough

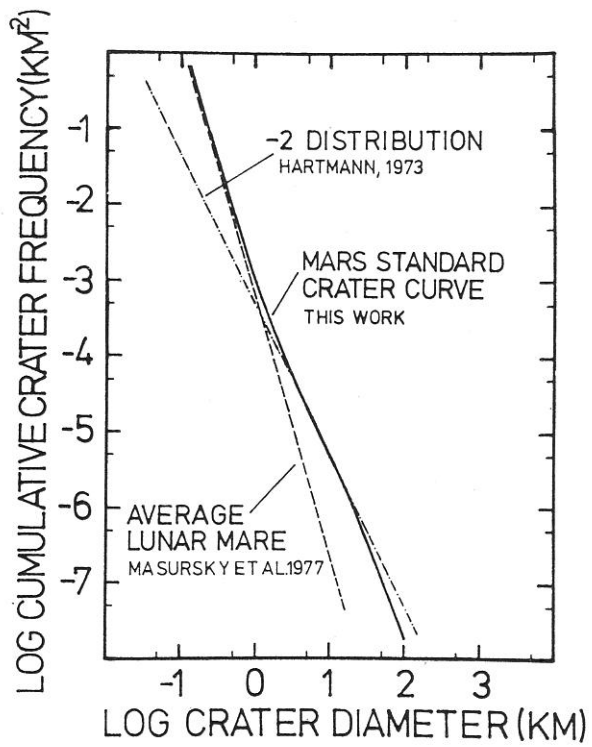


Fig. 2. Three Martian reference distributions. The D^{-2} distribution [*Hartmann,* 1973] and the distribution termed 'average lunar mare' [*Masursky et al.,* 1977] do not completely follow the actual distribution as measured by us [*Neukum and Wise,* 1976*a, b*] and extended here to diameters <1 km.

to decide whether this discrepancy is real or due to uncertainties in the determination of the functions over such a wide range of crater sizes.

Unfortunately, there is no unanimity in the use of reference production size-frequency distributions in the literature. This has rendered the comparison of reported Martian crater frequency data difficult and has sometimes led to very discrepant results. The situation is illustrated in Figure 2, where different Martian reference distributions proposed in the literature are compared. The D^{-2} distribution used by *Hartmann* [1973] and the distribution termed 'average lunar mare' used by *Masursky et al.* [1977] do not precisely follow the actual distribution as measured by us [*Neukum and Wise,* 1976*a*] and extended here to diameters <1 km. Application of those reference curves to measured Martian crater frequency data outside their ranges of approximate validity (where they deviate from our standard curve) would result in gross discrepancies.

Because the steepening of the Martian production size-frequency distribution near 2 km toward smaller diameters often has been questioned in scientific discussions, this observation deserves further elaboration. The steep distributions are measured on the younger Martian areas throughout as shown in the example of Figure 3. The data have been fitted with our Martian standard distribution showing reasonably good agreement. In fact, the steep distribution as measured by us at crater sizes $\lesssim$1 km has meanwhile been verified by others [e.g., *Masursky et al.,* 1977; *Dial,* 1978].

Some authors do not use such a fine diameter increment as we do in our cumulative or differential size-frequency distributions (18 steps per decade in a quasi-logarithmic distribution; see subsection of Data on Data Display) but use a $2^{1/2}$-diameter interval increment. Such a diameter increment is much too large for the determination of the change in slope around 1- to 2-km diameter, especially if measured down to sizes of about 0.5 km only, which is a common case.

Plescia and Saunders [1979] have introduced yet another reference distribution, namely, their counts in the Martian Lunae Planum region. We have replotted this distribution in

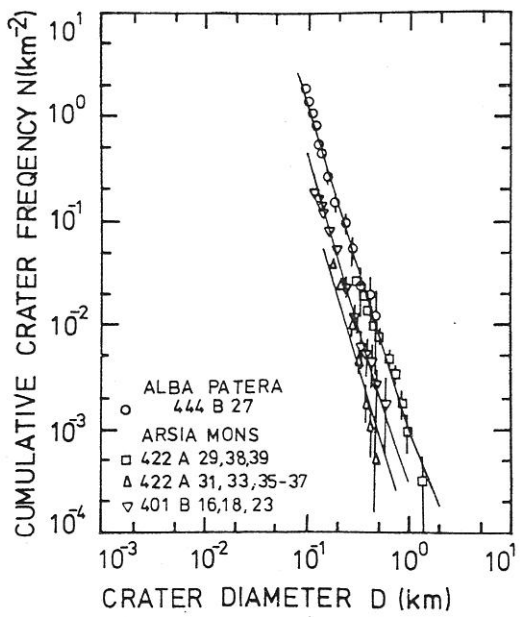


Fig. 3. Examples of Martian crater size-frequency data at sizes <1 km. The data have been fitted with our Martian standard distribution.

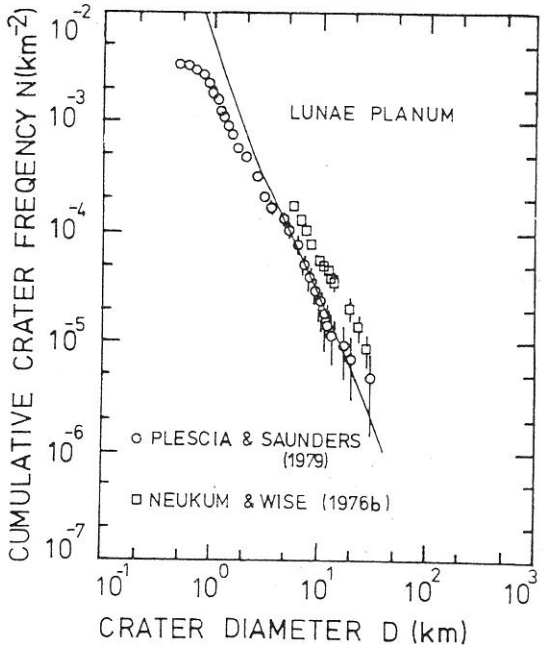


Fig. 4. Lunae Planum crater size-frequency distributions. The difference in crater density reflects counting in areas of differing age within Lunae Planum.

Figure 4 in comparison with *Neukum and Wise*'s [1976*b*] Lunae Planum counts and in comparison with the Martian standard curve. The Lunae Planum region had a very complex resurfacing history with preferential extinction of smaller craters by processes such as lava flooding and tectonic activities [cf. *Wise et al.,* 1979]. Such resurfacing activity is demonstrated by *Neukum and Wise*'s [1976*b*] measurements in a part of Lunae Planum showing the survival of large craters by their higher frequency (i.e., we look further back in time at those large crater sizes). Comparison of Plescia and Saunders' data with our Martian standard distribution shows that resurfacing has taken place also in their counting area and affected craters of <5-km diameter (deviation of their data from standard curve). On the basis of those arguments, Plescia and Saunders' Lunae Planum distribution is not an undistrubed production function but is affected by resurfacing processes. Therefore its use as a production size-frequency distribution or reference curve for determining relative ages can be questioned.

Extraction of Relative Ages From Martian Crater Size-Frequency Data

As discussed before, the extraction of relative ages (i.e., crater retention ages) from measured impact crater size-frequency data may be complicated because of resurfacing processes that selectively erased craters of certain sizes. (As discussed in the previous section, we neglect the effect of different target response possibly also to some extent responsible for irregularities in the size-frequency distributions.) We shall clarify our methodology of extraction of relative ages in the following discussions by giving four examples.

Lava flooding with destruction of small craters has been a common process on the Martian plains. The effect on crater populations is preferential destruction of small craters and the survival of larger ones. Depending on the magnitude of flows, on flow thicknesses, and frequency of occurrence, those effects are recorded in the crater populations in the way that the size-frequency distribution measured on such areas show irregu-

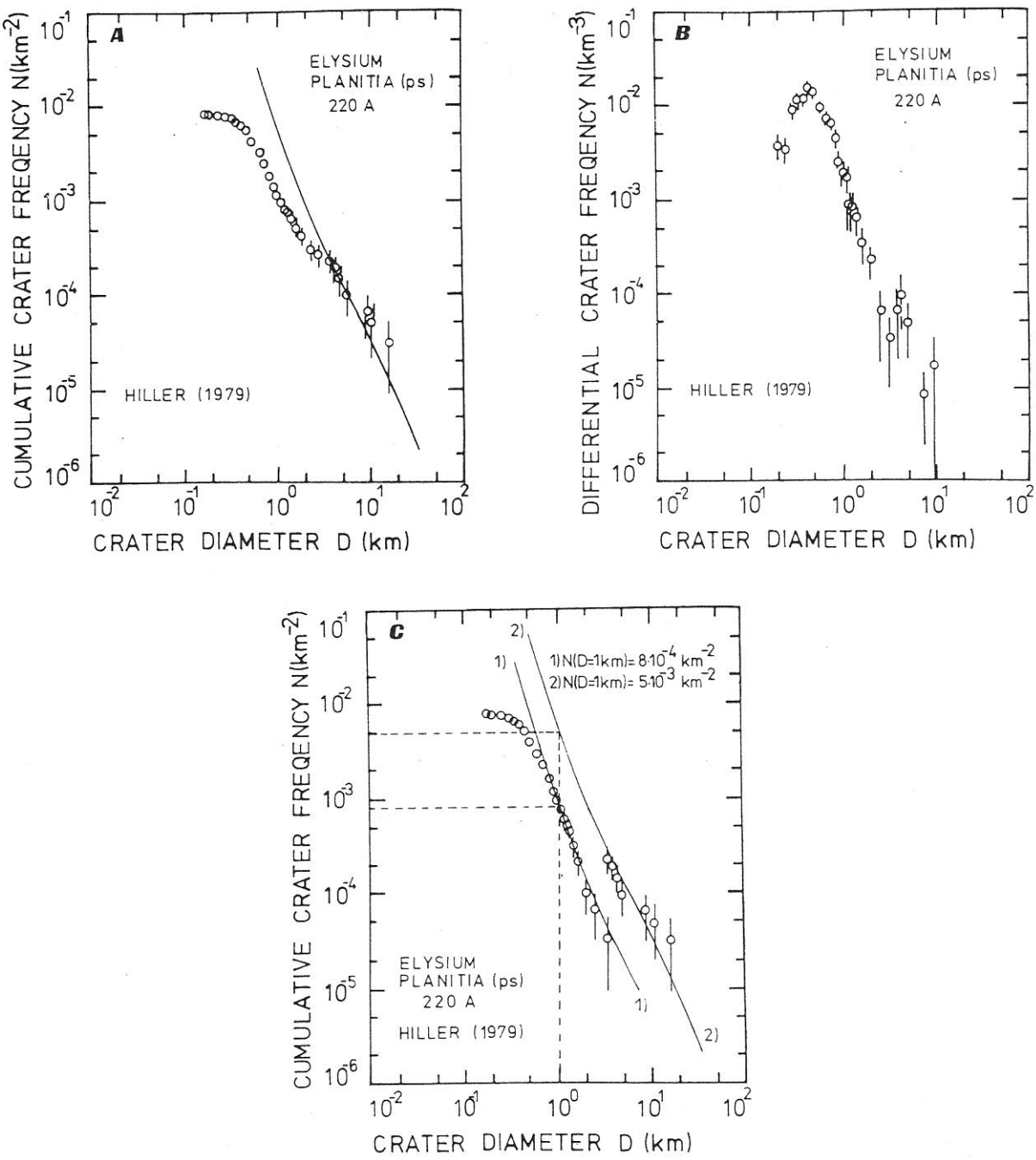


Fig. 5. Example of effect of resurfacing on the production size-frequency distribution in a plains area and extraction of relative ages. (*a*) Cumulative size-frequency distribution. (*b*) Differential size-frequency distribution. (*c*) Separation of the preresurfacing and the postresurfacing crater population.

larities in comparison with the undisturbed production size-frequency distribution, i.e., the standard distribution. The distributions may appear flattened at small sizes or may have irregularities at certain sizes. If one major flow occurred, then just one irregularity (bump) may be produced. An example of essentially a one-stage flooding history is given in Figure 5, displaying the frequency measurements in Elysium Planitia (smooth plains material) at about 20°N, 200°W (latitude, longitude). Comparison of the counts in Figure 5*a* with the standard distribution shows a deviation in crater frequency from the standard distribution at about 4-km crater diameter.

This is better shown in the differential distribution plot of the measurements displayed in Figure 5*b*. The differential distribution (the diameter derivative of the cumulative distribution or the approximation $\Delta N/\Delta D$) allows a more exact determination of the crater diameter below which lava flooding erased smaller craters. Craters smaller than 4 km have been almost completely erased in this area (decrease in differential frequency) and then the new surface has been cratered again (branch of curve where it steepens again at smaller sizes). Craters above $D = 4$ km survived flooding. The preresurfacing and the postresurfacing crater populations can approximately be separated as shown in Figure 5*c*. The cumulative distribution above the 'critical resurfacing diameter' $D = 4$ km gives

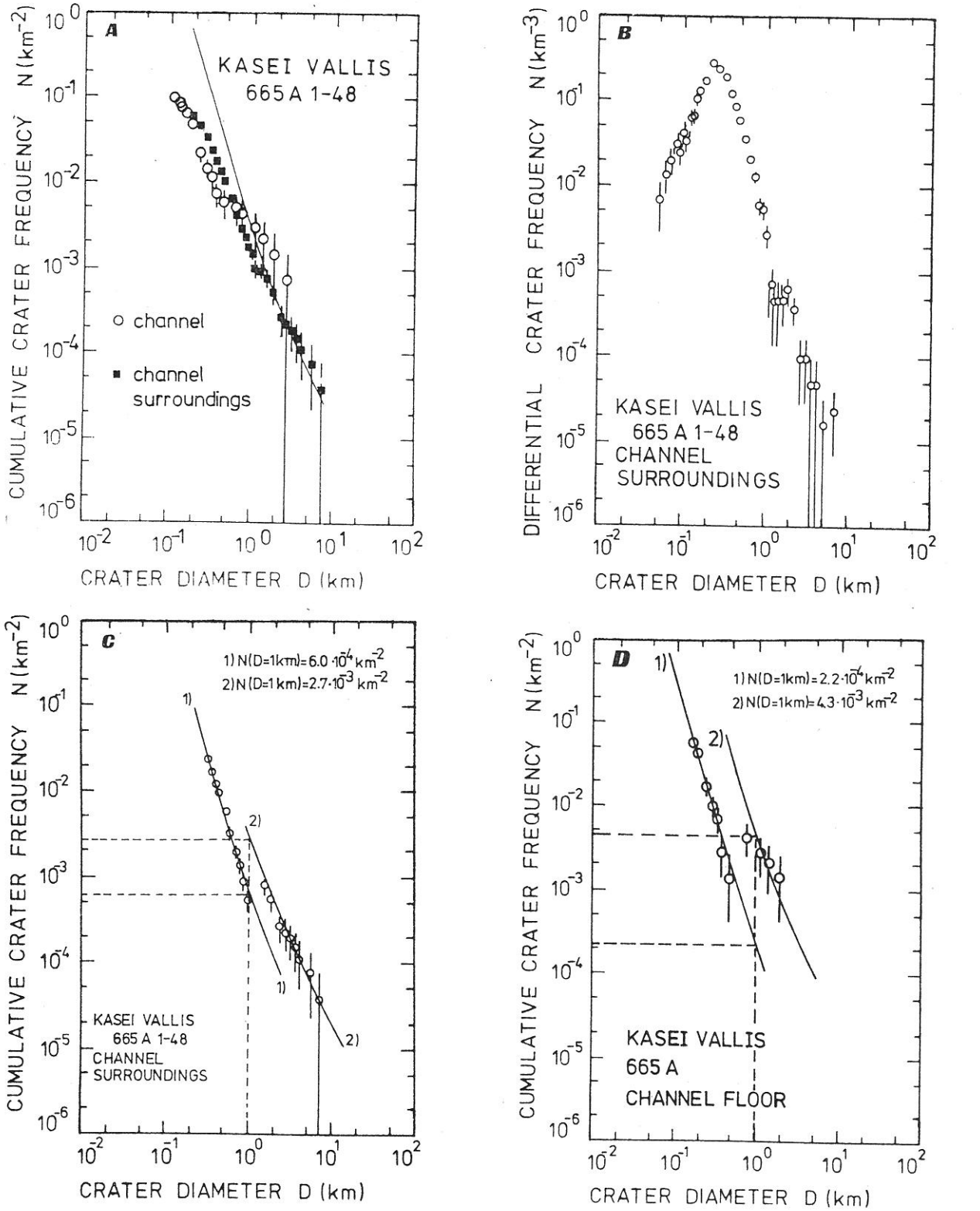


Fig. 6. Example of effect of resurfacing on the production size-frequency distribution in a channel area and extraction of relative ages. (*a*) Cumulative size-frequency distribution. (*b*) Differential size-frequency distribution (channel surroundings). (*c*) Separation of the pre-resurfacing and postresurfacing crater population of the channel surroundings and extraction of different crater retention ages at $D = 1$ km. (*d*) Separation of the preresurfacing crater populations of the channel floor and extraction of different crater retention ages at $D = 1$ km.

the crater retention age, $N(1)$, at $D = 1$ km through projection of the standard curve. Subtraction of the cumulative frequency at $D = 4$ km from the total distribution of Figure 5*a* approximately yields the branch of the distribution without survivors, i.e., the crater retention age of the new surface, $N(1)$, at $D = 1$ km through application of the standard curve (or directly reading the data at $D = 1$ km).

The bending over of the data at the smallest sizes as in Figure 5*a* around $D = 0.5$ km, which is seen in Figure 5*b* as a maximum of frequency at this size and drop-off toward

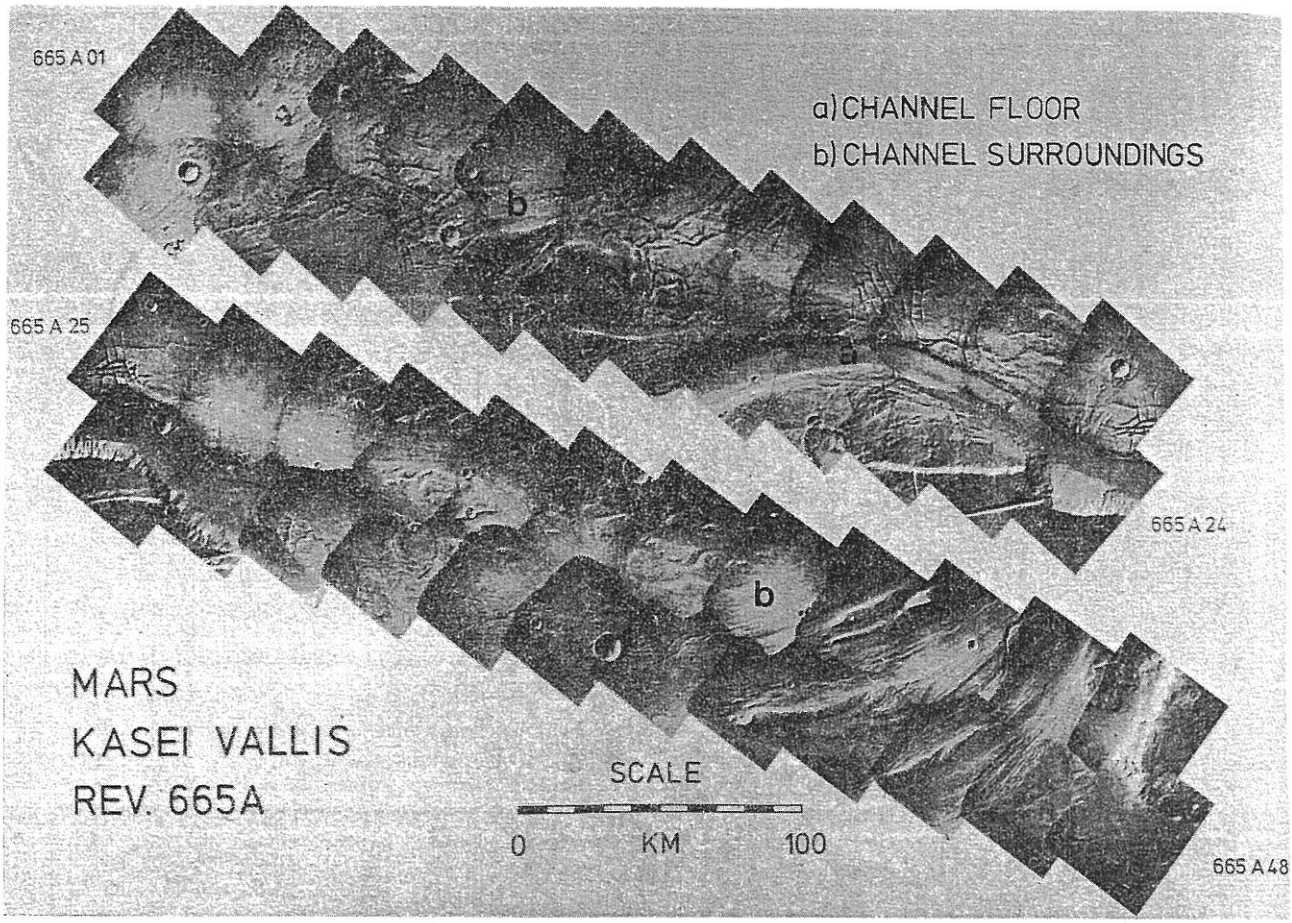


Fig. 7. Kasei Vallis (uncontrolled mosaic) at approximately 27°N, 67°W, area where the crater frequency data of Figure 6 have been obtained. The effects shown by the crater counts quantitatively can partially be seen in the surface morphology. Few large craters survived on the channel floor, seen as ghost craters, in some cases their rims just sticking out of the layer that smoothly covers the floor around them or in other cases completely covering them. Few small craters are present, in agreement with the counts. In the surroundings many craters of practically all sizes show traces of erosion; no clear-cut morphological difference as on the floor can be seen.

smaller sizes, reflects the effect of losing craters in counting near the resolution limit. In the following diagrams we have partially plotted the full counts also at the smallest sizes; in some cases, data have been truncated before bend over.

Examples of resurfaced crater populations in a channel area are given in Figure 6. The amount of resurfacing in channel areas has been found to be usually great with pronounced effects of flattening or bumps, as in these examples. Figure 6*a* displays the cumulative size-frequency distributions of counts in Kasei Vallis (ca. 27°N, 67°W) on the channel floor and in the channel surroundings. (The counting area is displayed in Figure 7). Comparison with the Martian standard distribution shows a flattening at sizes <2 km. Resurfacing has been greater on the channel floor than in the channel surroundings.

Though clearly shown in the cumulative distribution (Figure 6*a*), the differential distribution shown in Figure 6*b* reveals the effects of resurfacing. As discussed with the lava flooding effects in Elysium Planitia (Figure 5*b*), the kink of the distribution at $D = 2$ km marks the crater size below which erosion was effective and above which craters have largely not been destroyed.

Figure 6*c* gives the procedure of separation of the pre-resurfacing and post-resurfacing population of the channel surroundings and extraction of different crater retention ages at $D = 1$ km (cf. Figure 5*c*). Figure 6*d* illustrates the separation of the preresurfacing crater populations of the channel floor and extraction of different crater retention ages at $D = 1$ km. The effect of resurfacing at crater sizes <2 km has been greater on the channel floor than in the channel surroundings, whereas larger craters have survived better on the channel floor than in the surroundings. This suggests a different erosion mechanism.

Resurfacing effects on Martian volcanic structures are very common. The effects are very similar to those of lava flows on Martian plains. An example of a resurfaced crater population on a volcanic construct is given in Figure 8. We have used *Plescia and Saunders'* [1979] measurements for the Hecates Tholus volcano in order to demonstrate here our way of reinterpreting counts of other workers (whenever we could cast them into cumulative form again). Craters smaller than 3–4 km diameter are clearly deficient and are interpreted as the effects of resurfacing by lava flows (Figure 8*a*). The counts strongly deviate from the production distribution at those crater sizes and follow the production distribution closely at sizes <1-km diameter. A one-stage flooding of craters by lava is inferred. The preflooding and postflooding crater populations are separated in Figure 8*b* and the respective crater retention ages, $N(1)$, are given.

The preceding examples illustrate the technique of separating the different stages of resurfacing and permit tracing the

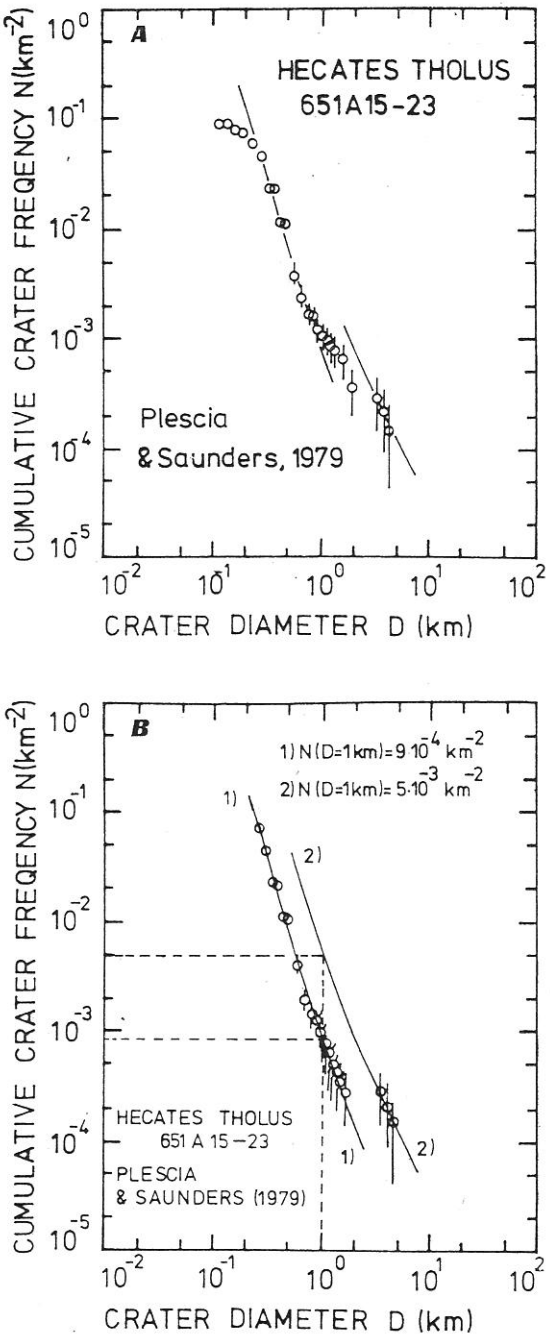


Fig. 8. Example of resurfacing by lava on a Martian volcano and extraction of relative ages. (*a*) Cumulative size-frequency distribution. (*b*) Separation of the preresurfacing and the postresurfacing crater populations and extraction of different crater retention ages at $D = 1$ km.

geologic development of a number of structures through Martian history.

Martian Cratering Chronology Models and Determination of Absolute Ages

In our definition, cratering chronology means the functional dependence of crater frequency on time in the past, i.e., on age. If this dependence is known, then in principle, it is possible to determine absolute ages for any planetary area from measured crater frequencies. Only in the case of the earth's moon has it been possible to determine the cratering chronology empirically by calibrating crater frequencies at (essentially) the Apollo landing sites with the radiometric ages of the respective lunar rocks [e.g., *Hartmann,* 1973; *Soderblom et al.,* 1974; *Neukum et al.,* 1975*b*]. In the case of all other planets, radiometric ages are unavailable and probably will remain unavailable for at least a decade.

Several attempts have been made to construct a Martian impact or cratering chronology on the basis of the lunar cratering chronology and extrapolating it to Mars. This has been done in different ways with different results. We will give a short summary in the following; for details the reader is referred to the work of *Hartmann* [1973, 1977, 1978], *Soderblom et al.* [1974], *Soderblom* [1977], and *Neukum and Wise* [1976*a, b*].

Neukum and Wise [1976*a, b*] compared the large-crater ($D \gtrsim 30$ km) frequencies on the lunar and Martian highlands and found similar crater frequencies, the Mars frequencies tending to be about a factor of 2 lower. This finding (in combination with the crater frequency/age relationship of the largest craters on Mars' moon Phobos) led to the proposal of similar impact rates and similar impact mass fluxes under the assumption of similar highland ages. Accounting for the lower impact velocities at Mars (about a factor of 1.5), a further conclusion was that the impact mass fluxes at the moon and Mars were close to equal. Since the Martian production size-frequency distribution was measured to be flatter than the lunar one, this led to 1-km crater frequency values for Mars a factor of 4.5 lower than the lunar ones for surfaces of identical age. Additionally, *Neukum and Wise* [1976*a, b*] proposed the same time dependence of impact flux for both planets (as did *Hartmann* [1973, 1977]). *Neukum et al.* [1978] reconsidered Neukum and Wise's data and came to the conclusion that 1-km crater frequency values for Mars should be only a factor of 3 lower than the lunar ones for surfaces of identical age. The other arguments have remained unchanged.

This relationship of the Martian to the lunar cratering chronology is given in Figure 9*a* (model I Martian cratering chronology). In the following, all relative ages have been converted by us to absolute ages by application of this Martian model cratering chronology. (In the following, absolute ages inferred from crater retention ages through application of cratering model chronologies are called 'cratering model ages.') It is, however, possible that for recent times ($\lesssim 3 \cdot 10^9$ years) the Martian cratering rate has been a factor of 2 higher than the lunar one [*Shoemaker,* 1977] whereas the ancient rates may have been nearly the same for 4- to 10-km diameter craters. The resulting Martian cratering chronology for 1-km cumulative crater frequency values is shown in Figure 9*b* (model II Martian cratering chronology). For old ages ($>3.5 \cdot 10^9$ years) it is identical to the Martian chronology of Figure 9*a*, and for ages less than $3 \cdot 10^9$ years it is a factor of 2 higher than the curve in Figure 9*a* but a factor of 1.5 lower than the lunar curve at those crater sizes. For larger crater sizes the ratio between lunar and Martian cratering chronology values is different owing to the different standard distributions; the lunar and Martian curves of Figure 9*a* would fall nearly upon each other at 4- to 10-km crater sizes (see Figure 10 and relevant discussions).

Soderblom et al. [1974] and *Hartmann* [1977] have proposed substantially different Martian cratering chronologies. This has been discussed by *Neukum and Wise* [1976*a*] and for clarification of some details by *Neukum et al.* [1978]. The arguments and facts given in those papers need additional clarification.

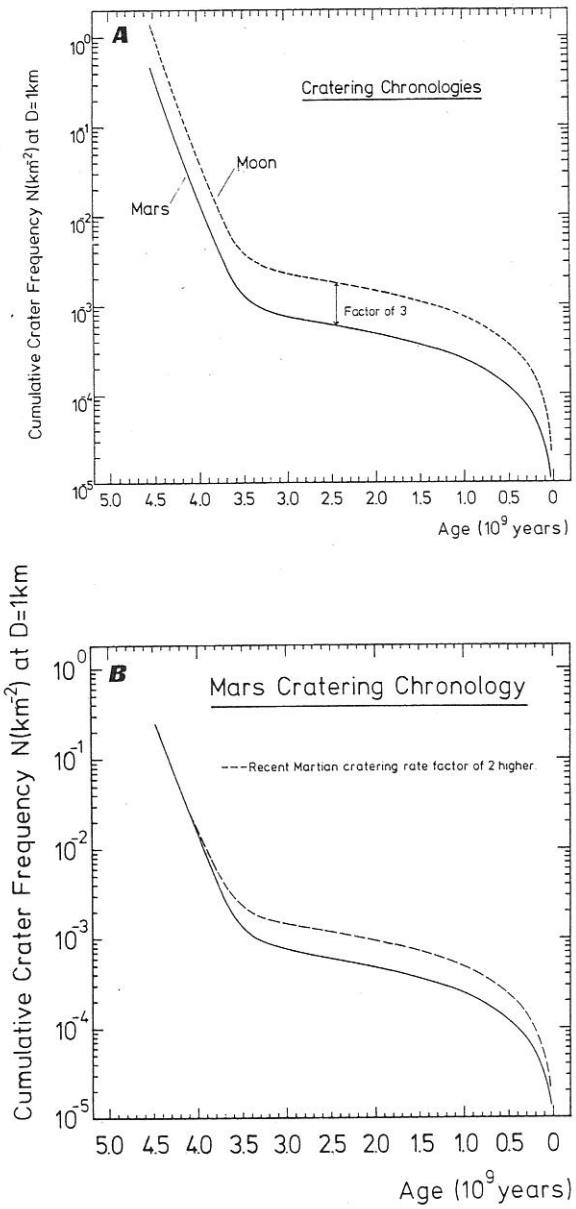


Fig. 9. Comparison of lunar and Martian cratering chronologies. (*a*) Cratering chronology model I: this 'model I' Martian cratering chronology is used throughout the paper for inference of cratering model ages from crater frequency data if not stated otherwise. (*b*) Cratering chronology model II: this chronology compares the model I Martian cratering chronology of Figure 9*a* with a possible cratering chronology (model II) with factor of 2 higher cratering rate at recent times ($<3 \cdot 10^9$ years).

A problem in comparing data from different authors has been the interpolation to other crater sizes. In order to minimize any bias, we have chosen to compare all three currently proposed Martian impact chronologies at 4- to 10-km sizes as shown in Figures 10*a* and 10*b*. Our Martian impact chronology has been reduced to crater frequencies for 4- to 10-km-diameter craters (equivalent to the difference of the cumulative crater frequencies at $D = 4$ km and $D = 10$ km). The solid line is a direct conversion to 4- to 10-km values of the Mars cratering chronology of Figure 9*a*. The dashed curve is based on the upper chronology of Figure 9*b* (model II chronology). The solid curve is nearly identical with our lunar cratering chronology curve for the 4- to 10-km crater size range (equivalent to being a factor of 3 lower than the lunar chronology curve at 1-km crater size (cf. Figure 9*a*)). We believe that the range between our lower curve (solid line) and our upper curve is the range of uncertainty in our knowledge of the Martian impact chronology.

The distribution from *Soderblom et al.* [1974] in Figure 10 is the original unconverted data as given in their graph. Hartmann's chronology curve is directly taken from his graph [*Hartmann,* 1973] giving 'adjusted crater densities' (normalized to his average lunar mare value and valid for the 4- to 10-km size range) and converted to Martian impact conditions by using his newest ratio of 2 [*Hartmann,* 1977] between Martian and lunar cratering rates. *Hartmann* [1977] changed his former [*Hartmann,* 1973] Martian cratering chronology drastically by assuming a new ratio of 2 instead of 6 between Martian and lunar cratering rates.

Consideration of the total number of craters between 4 km and 10 km as shown in Figure 10 has the advantage of minimizing the effects of assumptions required to transfer different data sets to a uniform size. *Neukum and Wise* [1976*a, b*] inconsistently reduced *Soderblom et al.*'s [1974] data to 1-km values by application of the lunar crater size-frequency production curve rather than the Martian one. This inconsistency was removed by *Neukum et al.* [1978] by applying the proper distribution. Here, Soderblom et al.'s and Hartmann's data are practically their original data determined mainly in the 4- to 10-km range. The reduction of our data, which were originally given as numbers at $D = 1$ km to this size range, does not introduce any bias, since it applies our Martian standard distribution curve as also used for conversion to $N(1)$ numbers from all larger (or smaller) crater sizes.

Martian chronologies proposed by Soderblom et al. and Hartmann differ significantly from ours. If we apply the three chronologies to the same Martian crater frequency data of, e.g., $N(4\text{–}10) = 100$ (10^{-6} km^{-2}), we would read an absolute age of about $3.7 \cdot 10^9$ years from our chronology but absolute ages of $1.4 \cdot 10^9$ years and $1.2 \cdot 10^9$ years from Hartmann's and Soderblom et al.'s chronology, respectively. This represents a difference of more than $2 \cdot 10^9$ years. For larger crater frequencies the differences would be smaller. One reason for such differences was that the discrepancies in the Martian model chronologies just reflect discrepancies in the lunar chronologies that they are based on or are referred to [*Neukum and Wise,* 1976*a*]. An error in Soderblom et al.'s lunar data has already been pointed out (an erroneous conversion of the lunar data, which originally were measured in terms of an erosion parameter (D_L values [*Soderblom and Boyce,* 1972]) to $N(4\text{–}10)$ crater frequency values; the relevant data are given in Table 1). *Soderblom* [1977] corrected his data but without explicitly giving the lunar and Martian chronology curves again. He maintained that (1) the Martian cratering rate was about a factor of 1.5 higher than the lunar one in the last $4 \cdot 10^9$ years and (2) his Martian chronology was determined independently of the lunar one in the 5- to 10-km size range in assuming (as did *Soderblom et al.* [1974]) 'that the oldest plains on Mars record the accumulation of impact craters in this size range over the last $4 \cdot 10^9$ years.'

If the Martian cratering rate was about a factor of 1.5 higher than the lunar rate in the last $4 \cdot 10^9$ years, his Martian cratering chronology should fall nearly on ours if both data sets are correct. As shown in Figure 10, however, *Soderblom et al.*'s [1974] data differ by a factor of >5. In his recent paper [*Soderblom,* 1977] he has not explicitly given the Martian cratering chronology but a crater frequency value for 5- to 10-km

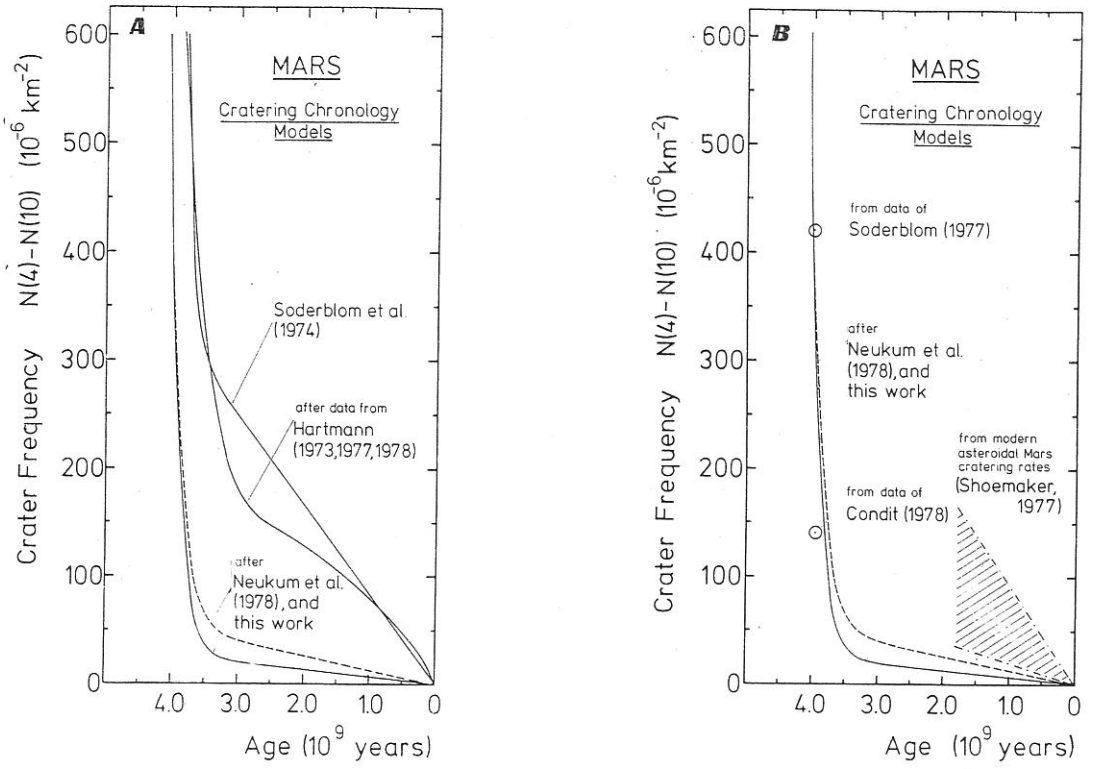


Fig. 10. Comparison of proposed Martian impact chronologies. (*a*) Our Martian impact chronology (Figure 9*b*, model I chronology, solid line; model II chronology, dashed line) has been reduced to crater frequencies for 4- to 10-km-diameter craters (equivalent to the difference of the cumulative crater frequencies at $D = 4$ km and $D = 10$ km). The data from *Soderblom et al.* [1974] are the original unconverted ones, as given in his graph. Hartmann's chronology curve is directly taken from his graph [*Hartmann,* 1973] giving 'adjusted crater densities' (normalized to his average lunar mare value) and converted to Martian impact conditions by using his newest ratio of 2 between Martian and lunar cratering rates. (*b*) Comparison of our Martian impact chronology (model I and II) with *Soderblom*'s [1977] and *Condit*'s [1978] new data for the oldest (ca. 4 b.y.) Martian plains and with data from modern asteroidal impact rates.

TABLE 1. Comparison of Lunar Cratering Data

	Age, 10^9 years	Crater Frequency	Hartmann	Soderblom	Neukum	Shoemaker
Average mare	3.4[a]	$N(4–10)$	160[b]			
Mare Serenitatis		$N(4–10)$	77[c] (60)		50[d]	
		$N(10)$			ca. 10[d]	
$3.3 \cdot 10^9$ year		$N(4–10)$	56[e]	30[f], 210[g]	30[d]	100[h]
old mare surface		$N(10)$				20[h]
Orientale Basin	ca. 3.9[i]	$N(4–10)$	380[c]	70[f], 390[g]	250[d]	
		$N(10)$			ca. 60[d]	
Copernicus	0.8–1[j]	$N(4–10)$	48[k]	8[f], 70[g]		
Fra Mauro	3.9–4[l]	$N(4–10)$	400[m]	230[f], 570[g]	340[d]	

$N(4)$–$N(10) = N(4–10)$ means crater frequency in the diameter interval $D = 4$ to 10 km, which is identical with the difference of the cumulative crater frequencies at $D = 4$ km, $N(4)$, and at $D = 10$ km, $N(10)$. Only original data, directly given in the size range 4 to 10 km as numbers or in diagrams, have been used. Data were originally given in cumulative or incremental form by the authors exclusively for either 4 km or 10 km or directly in the form $N(4–10)$. For conversion, either the law $N \sim D^{-2}$ has been used or our lunar standard distribution curve, which gives identical results within a few percent. Crater frequency data are given per 10^6 km^2. The names Hartmann, Soderblom, Neukum, and Shoemaker stand for usually a larger group of co-authors.

[a] As given by *Hartmann* [1973, 1977].

[b] From *Hartmann* [1973, 1977, 1978]: his average mare value, which he gives in relative frequency numbers as 1, really varies somewhat in absolute frequency throughout his papers.

[c] *Hartmann* [1967]: $N(4–10) = 77$ (10^{-6} km^{-2}) is an average value from his D^{-2} fit. Really measured at 4–10 km is $N(4–10) = 60$ (10^{-6} km^{-2}), nearly identical with the measurement by *Neukum et al.* [1975*a*].

[d] *Neukum et al.* [1975*a*, *b*].

[e] *Hartmann* [1973]: calculated from his crater frequency value relative to average mare, 0.35, and his average mare value of $N(4–10) = 160$ (10^{-6} km^{-2}) (see footnote *b*).

[f] *Soderblom et al.*'s [1974] data corrected by us for the conversion error (cf. footnote *g*) by use of the relationship between D_L values and crater frequencies form *Neukum* [1977] and *Boyce and Johnson* [1978], and by use of the original data of *Soderblom and Boyce* [1972] (average values).

[g] From *Soderblom et al.* [1974]; value for Cayley formation considered contemporaneous with Orientale by the authors. These data were erroneously determined by *Soderblom et al.* [1974] (from D_L values [*Soderblom and Boyce,* 1972]) as admitted by *Soderblom* [1977].

[h] *Shoemaker* [1977], estimated for $D = 10$ km within ±50%.

[i] Inferred by a variety of authors comparing crater count data in combination with the argument that the oldest mare basalt ages are about $3.9 \cdot 10^9$ years and Orientale is the youngest basin, i.e., younger than the Imbrium Basin, which means $4 \cdot 10^9$ years old.

[j] Compared, for example, *Eberhardt et al.* [1973] or *Alexander et al.* [1976].

[k] *Hartmann* [1973]: calculated from his frequency value relative to average mare, 0.3, and his average mare value of 160 (10^{-6} km^{-2}) (see footnote *b*).

[l] Compare, for example, *Tera et al.* [1974].

[m] *Hartmann* [1973]: from his Fra Mauro/average mare ratio = 2.5, nearly identical with his Orientale basin event/average mare ratio of 2.4.

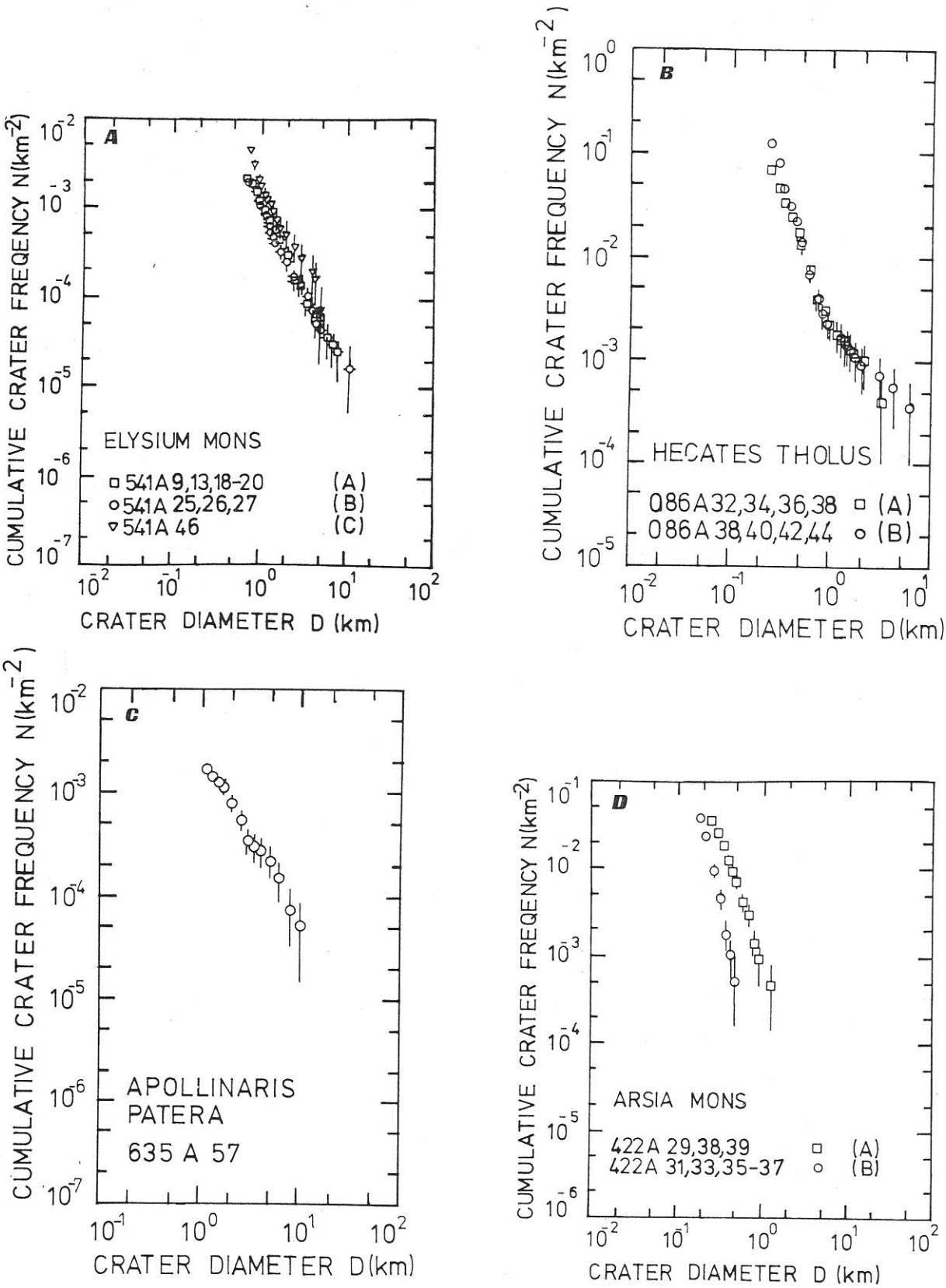


Fig. 11. Cumulative crater size-frequency data of our measurements on Martian volcanoes. The respective 1-km-diameter crater frequency values (retention ages $N(1)$) are given in Table 2.

craters and an age of $4 \cdot 10^9$ years, $N(5)$–$N(10) \approx 240$ (10^{-6} km^{-2}), which corresponds to $N(4)$–$N(10) \approx 420$ (10^{-6} km^{-2}). This point falls right on our Martian cratering chronology (Figure 10*b*). There seems to be a problem of determining the crater frequencies of the oldest Martian plains and assigning the plains ages. We shall discuss this problem in greater detail.

Soderblom et al. [1974] have given the frequency of 4- to 10-km craters on the oldest plains such as Lunae Planum as $N(4$–

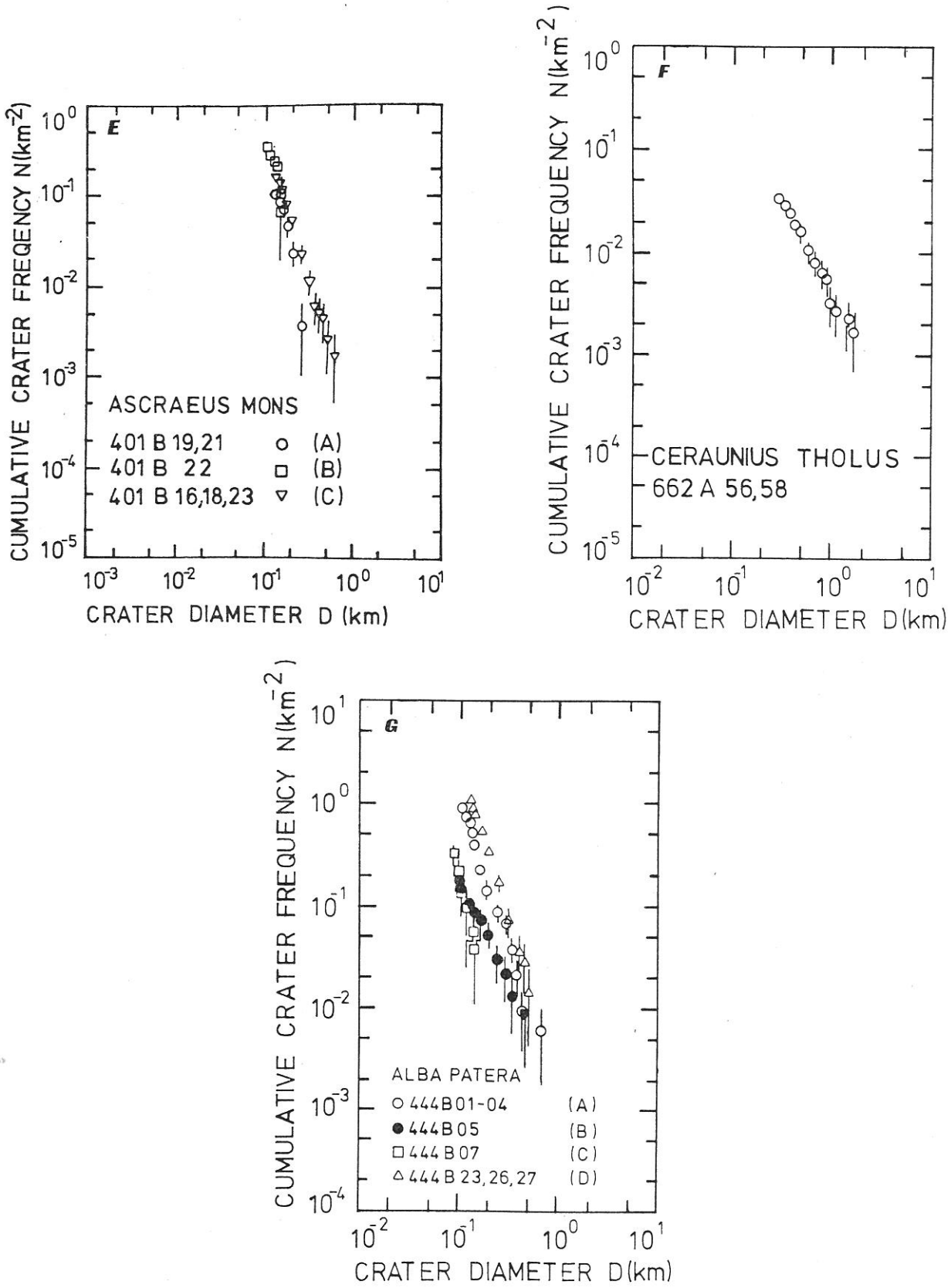


Fig. 11. (continued)

10) = 130–180 (10^{-6} km^{-2}) and values of $N(4–10)$ = 200–250 (10^{-6} km^{-2}) for part of the Amenthes uplands judged by them to record the ancient crust just after the early intense bombardment. Another value given for the oldest plains and the age of $4 \cdot 10^9$ years is 240 per 10^6 km^2 for 5- to 10-km craters [*Soderblom*, 1977], which is equivalent to $N(4–10) = 420$ (10^{-6} km^{-2}).

We read from *Soderblom et al.*'s [1974] diagram a value of $N(4–10) \approx 250$ craters per 10^6 km^2 for a 3-b.y.-old surface, not a 4-b.y.-old one. His plot does not allow reading a value at

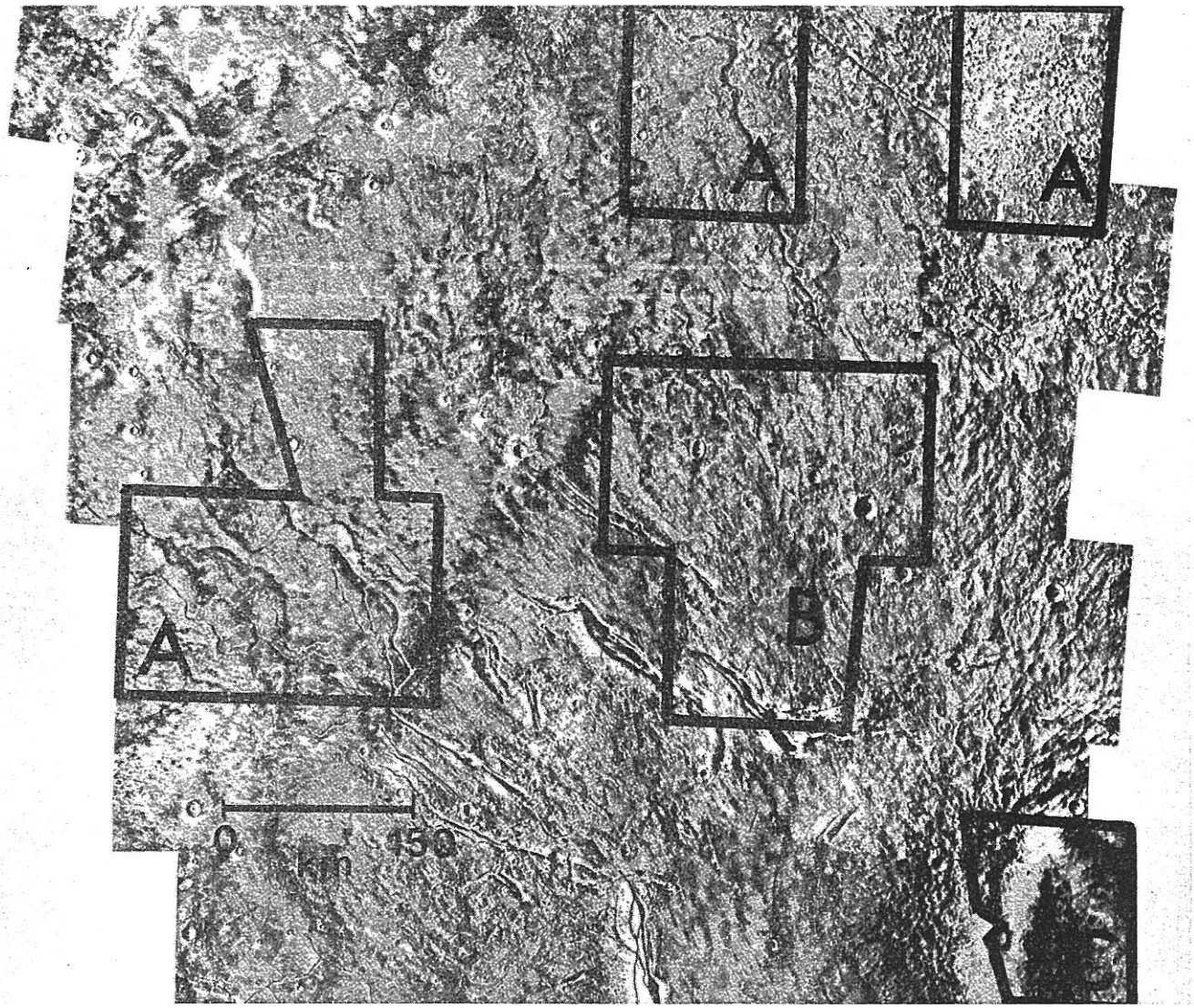


Fig. 12*a*. Elysium Mons with areas selected for crater counts: shield without caldera (C); aureole excluding Elysium Fossae (B); surrounding rolling plains (A) material (Figure 11*a*).

$4 \cdot 10^9$ years but at about 3.8 b.y., which gives a frequency of $N(4\text{–}10) = 600$ (10^{-6} km^{-2}), i.e., $N(4\text{–}10) > 600(10^{-6}$ km$^{-2})$ for a 4-b.y.-old Martian surface. Thus his own data appear to be inconsistent. If we consider just the crater frequency for the oldest plains (Lunae Planum), Soderblom et al.'s data cover a factor of 3 from $N(4\text{–}10) = 130$ (10^{-6} km^{-2}) to $N(4\text{–}10) = 420$ (10^{-6} km^{-2}). The data of Soderblom and co-workers become even more discrepant if all the crater frequency values they give for a $4 \cdot 10^9$ year old surface regardless of geologic unit are taken. In fact, the crater frequency of $N(4\text{–}10) = 130\text{–}180$ (10^{-6} km^{-2}) for Lunae Planum is assigned a $2 \cdot 10^9$ year age by *Soderblom et al.* [1974], whereas the part of Amenthes uplands believed by the authors to date back to the time right after the early intense bombardment (ca. $4 \cdot 10^9$ years ago) and showing crater frequencies of $N(4\text{–}10) \approx 200\text{–}250$ (10^{-6} km^{-2}), is assigned an age of about $3 \cdot 10^9$ years. *Soderblom* [1977], however, assigns an age of $4 \cdot 10^9$ years to the frequency of $N(4\text{–}10) = 420$ (10^{-6} km^{-2}), whereas *Soderblom et al.*'s [1974] data of $N(4\text{–}10) = 600$ for ca. $3.8 \cdot 10^9$ years would lead to a value of $N(4\text{–}10) > 600$ (10^{-6} km^{-2}) for an age of $4 \cdot 10^9$ years.

Recently, *Condit* [1978] redetermined the crater frequency on the oldest Martian plains, obtaining a value of $N(4)\text{–}N(10) \approx 140$ (10^{-6} km^{-2}). If we correlate this number with the age of ca. $4 \cdot 10^9$ years suggested by *Soderblom* [1977], we obtain a data point falling right below our Martian cratering chronology. We conclude that *Soderblom*'s [1977] approach and *Condit*'s [1978] data roughly confirm our Martian cratering chronology for ages of ca. $4 \cdot 10^9$ years. Between 4 and 3 b.y. ago the lunar cratering rate still decreased about 1 order of magnitude. If the Martian cratering rate dependence was similar, we would conclude from the close agreement of the data at ca. $4 \cdot 10^9$ years that the Martian cratering chronology that could be developed from their data is coincident with our model cratering chronology within an uncertainty of less than a factor of 2.

Crater frequencies calculated from *Shoemaker*'s [1977] modern asteroidal Mars cratering rate data for the last $2 \cdot 10^9$ years would fall higher than our Martian cratering chronologies as given in Figure 10*b*. The uncertainties in the determination of the absolute impact rate on Mars from the observed asteroid abundances is rather high, however, and the actual current impact rate may lie much lower. The ratio of asteroidal cratering rates at the Moon and Mars of 1:2 given by *Shoemaker* [1977] is considered much more reliable than

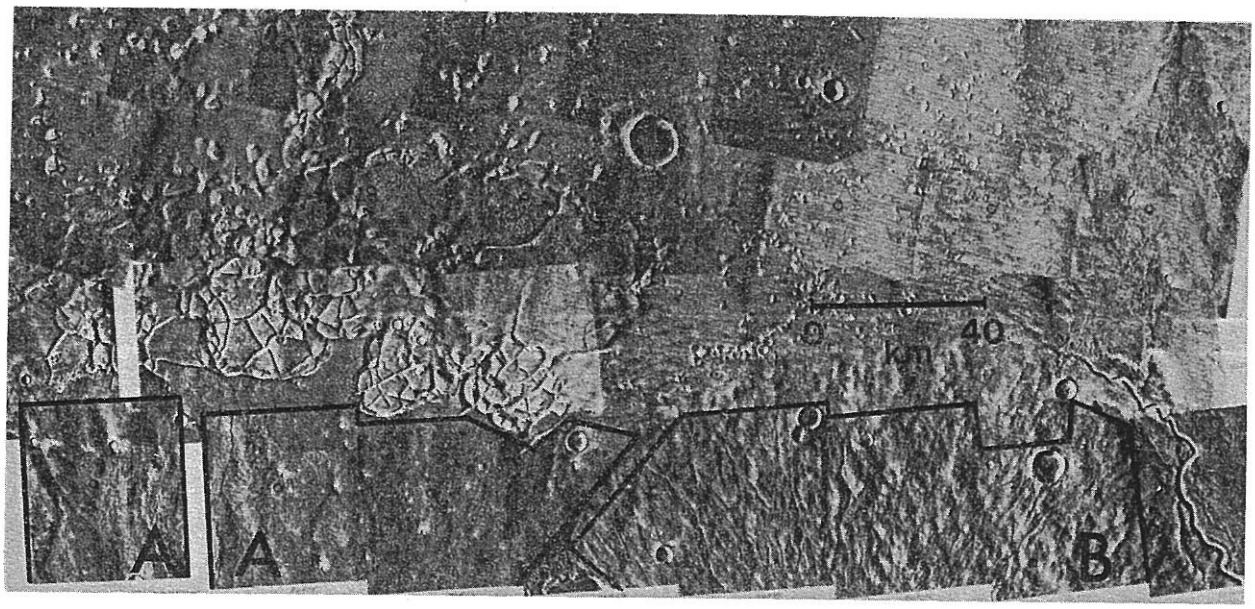


Fig. 12*b*. Hecates Tholus with areas selected for crater counts: part of the shield (B); surroundings with lava flow surface (A) (Figure 11*b*).

the absolute values for either planet. There seems to be a disagreement in the lunar asteroidal impact rates, however, in the scientific community. *Kresak* [1980] would argue for a total abundance of earth-crossing asteroids (which could impact the moon) being about a factor of 2–3 lower than Shoemaker's value, thus leading to lower cratering rates in the same proportion. If that is true, the asteroidal data would be in reasonable agreement with our lunar and Martian cratering chronologies for recent times.

Hartmann's lunar and Martian cratering chronologies also lie systematically higher than our chronology (Figure 10*a*). This may be due to his discrepant lunar data. Although his

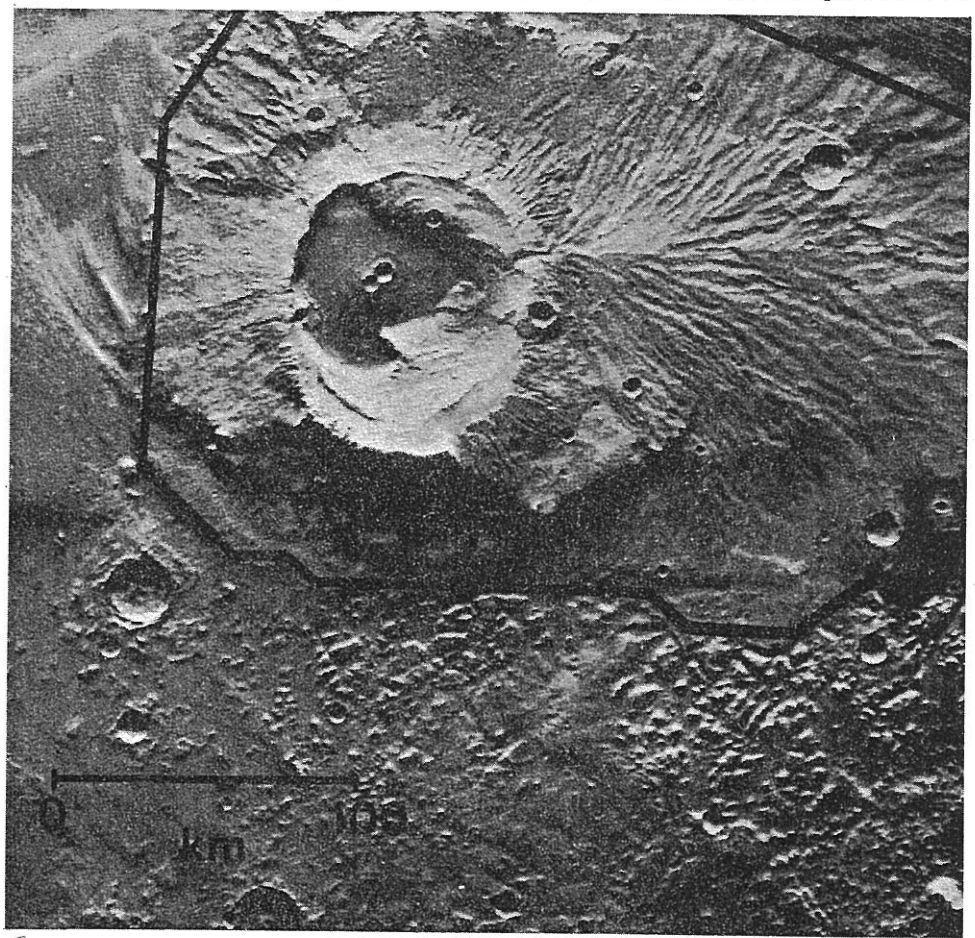


Fig. 12*c*. Apollinaris Patera with areas selected for crater counts: the shield of this volcano shows a young lava fan; measurements include the caldera (Figure 11*c*).

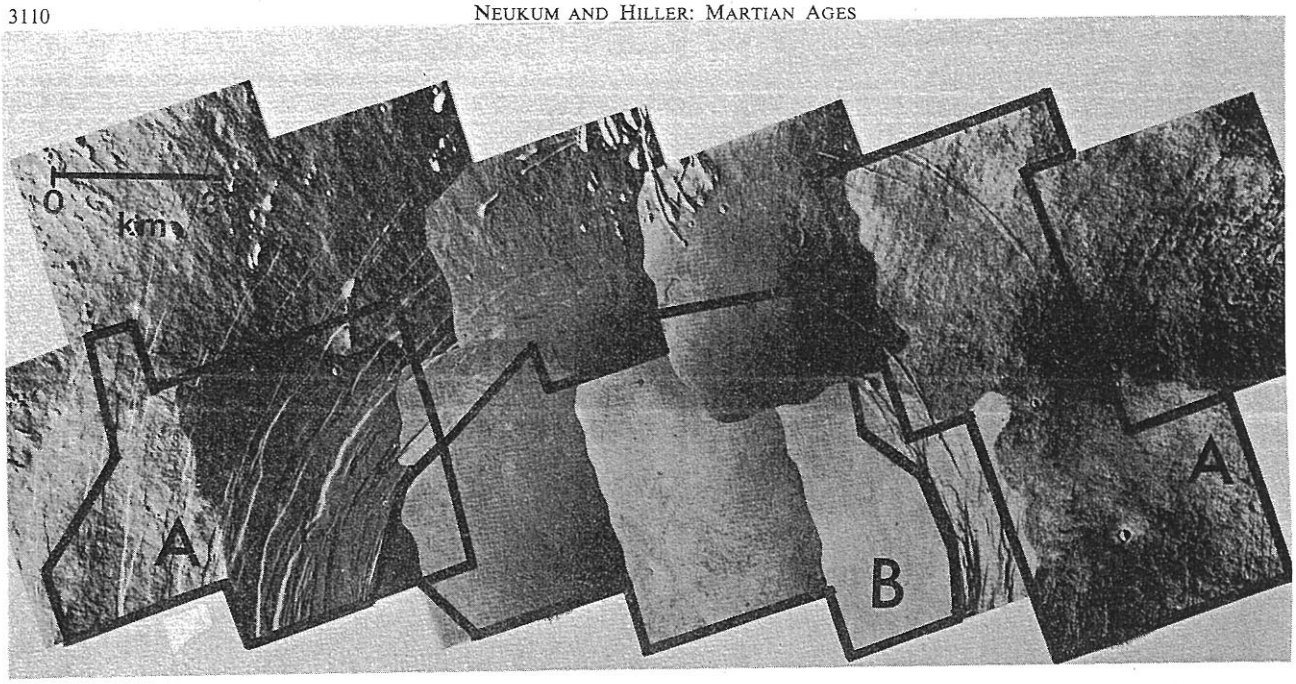


Fig. 12*d*. Arsia Mons with areas selected for crater counts: innermost undisturbed floor of the caldera (B); adjacent flanks (A) (Figure 11*d*).

average mare value for an age of $3.4 \cdot 10^9$ years is too high (Table 1), his values for a $3.3 \cdot 10^9$ year old mare surface is discrepant by less than a factor of 2 and is almost identical for Mare Serenitatis. Thus his lunar cratering chronology should lie within a factor of <2 of our lunar cratering chronology; likewise, his Martian cratering chronology should lie less than a factor of 2 higher than our model II Martian chronology in Figure 10 (since he now assumes a Martian to lunar cratering rate ratio of 2 [*Hartmann,* 1977]). However, Hartmann's chronology appears to be tied to relatively poor data of Copernicus or the frequency of craters on the terrestrial Canadian shield. Those data yield very high crater frequency numbers. For example, his $N(4\text{–}10)$ value for Copernicus (probable age $\approx$ 900 m.y.) is about as high as the $N(4\text{–}10)$ value for a $3.3 \cdot 10^9$ year old lunar surface or for Mare Serenitatis (Table 1). In addition, his crater frequency curve for the maintained constant flux in the age range $<3 \cdot 10^9$ years [*Hartmann,* 1973] is not consistent with a constant flux behavior.

Consequently, as *Neukum and Wise* [1976*a*] pointed out, at least part of the reasons for the discrepant Martian chronologies of different workers is the discrepancy in the lunar cratering chronologies. A profound reexamination of the lunar data and the techniques used in data reduction are necessary in our opinion. Our data have the advantage that we have used the same system of data reduction throughout and the data are at least internally consistent.

An important factor in data reduction is the use of common production size-frequency reference curves. For example, we believe that part of the differences between Hartmann's and our data reflects Hartmann's omission of the steepening of the crater size-frequency production function near 1–2 km and maintaining a $N \sim D^{-2}$ production function throughout. Ap-

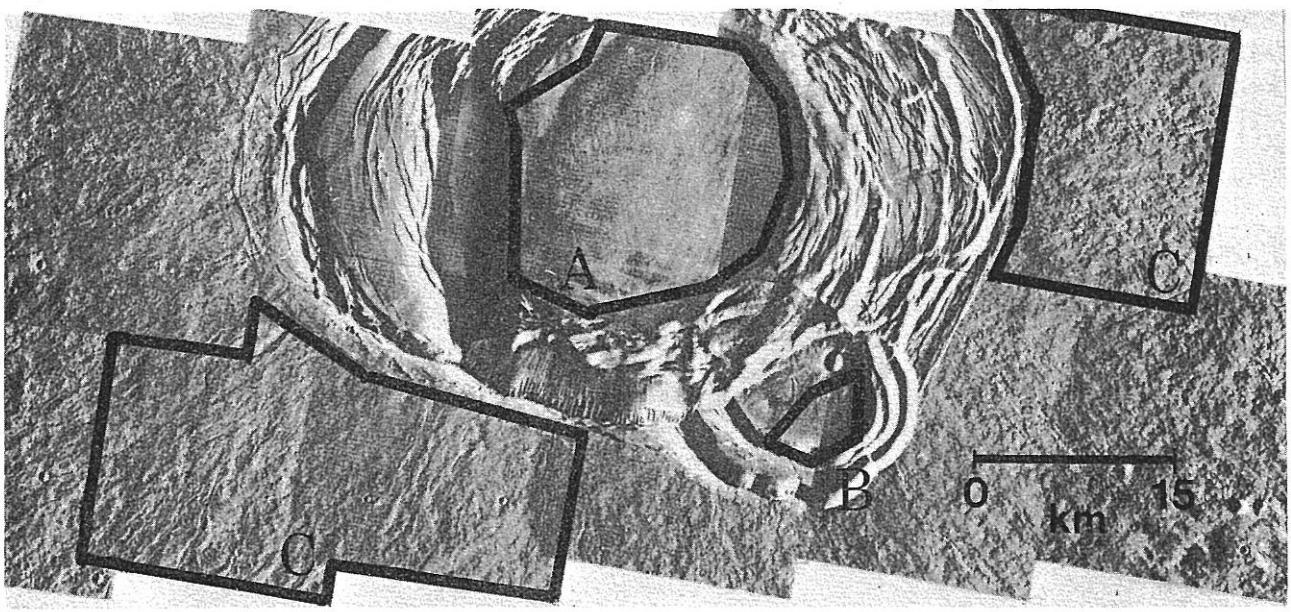


Fig. 12*e*. Ascraeus Mons with areas selected for crater counts: two areas on the flanks (C); three measurements on the floor of different parts of the caldera (A, B) (Figure 11*e*).

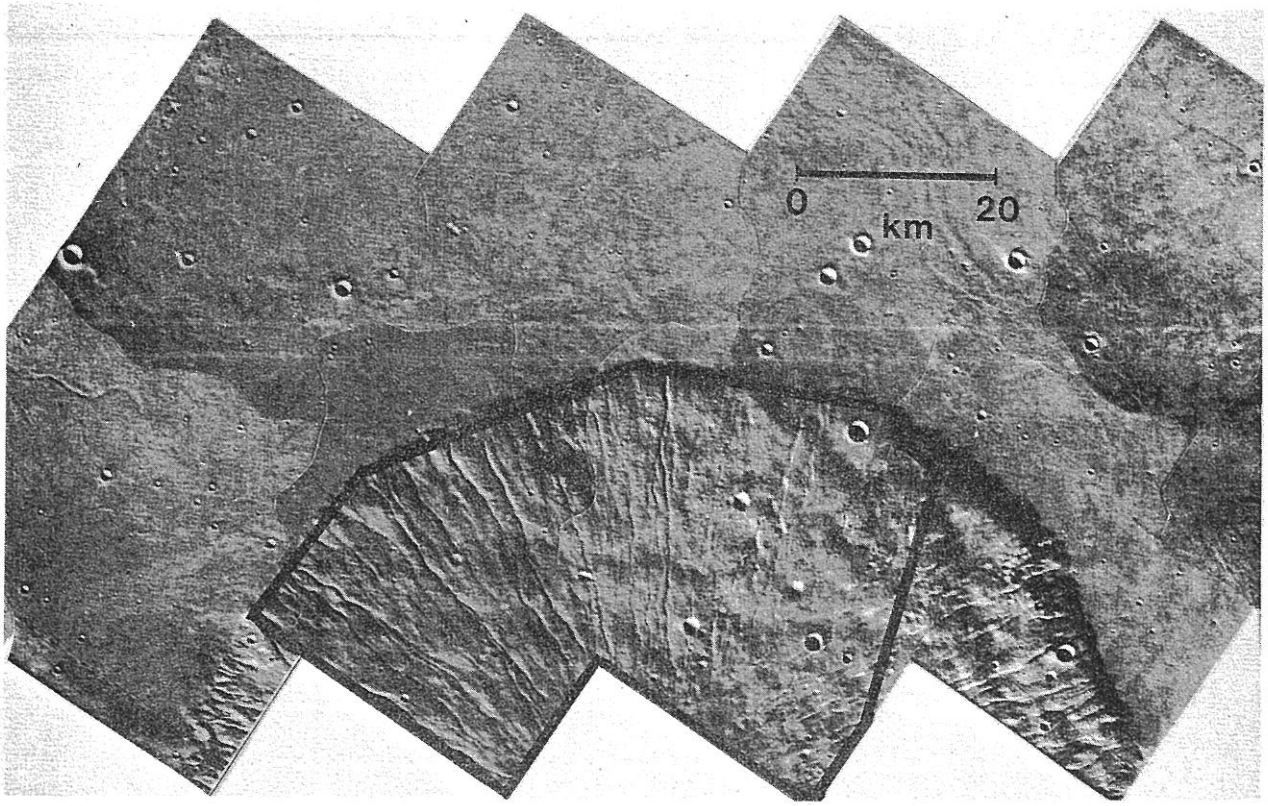


Fig. 12*f*. Ceraunius Tholus with areas selected for crater counts: one half of the shield (Figure 11*f*).

plications of such a production function to the crater populations at some Apollo landing sites where only craters smaller than 1–2 diameters can be measured (e.g., Apollo 15, Apollo 12) results in crater frequency values in the 4- to 10-km size range that are too high. However, this is the size of craters that he emphasizes.

DATA

This section consolidates crater frequency data from our previous work, published data from other authors, and newly acquired data. Every effort has been made to reduce the data in the same way. The procedure of measurement preparation and data evaluation is summarized below.

Preparation of Measurement Area

Photogeologic interpretations from uncontrolled mosaics defined homogeneous units. Terminology of photogeologic units is based on the Geologic Map of Mars, scale 1:25,000,000, by *Scott and Carr* [1978]. In positive transparencies of single Viking or in some cases Mariner 9 images, the various areas selected for measurement have been outlined by polygons.

Secondary craters or craters of nonimpact origin have been identified and marked for exclusion in the measurements as similarly done in our lunar work [*Neukum et al.*, 1975*a*] or as done by *Blasius* [1976]. The effect of few possibly undetected random secondaries on the size-frequency distributions is considered negligible.

Data Acquisition

Our measurements were performed with a high-precision Zeiss stereo comparator (for details, see *Neukum et al.* [1975*a*]). Data were cast by computer processing into special formats allowing listings of relevant parameters of measurement such as area names and sizes, crater numbers, cumulative, differential, or relative size frequencies, etc. or graphical displays of those data, and allowing interactive interpretation with a Hewlett Packard/Kontron computer-digitizer system.

The treatment of data from other workers depended on the form of publication. In the case of availability of listings displaying all relevant parameters, data were converted into our system (cf. the next two subsections) directly. If data were given in crater size-frequency distribution curves, the curves were redigitized with the Hewlett-Packard/Kontron computer-digitizer system (0.1-mm accuracy). Error bars and size of measurement areas were recalculated if possible. This method, although not totally accurate, gives adequate results and provides the only chance to compare crater count data displayed in varying forms.

Data Reduction and Interpretation

Data reduction and interpretation were performed with the interactive Hewlett Packard/Kontron digitizer-computer system. As discussed before, cumulative crater size-frequency data were fitted with the Martian standard distribution. In cases of several phases of surface emplacement documented in the data, parts of the data representing underlying and therefore older strata were subtracted to separate the different units. Crater retention ages in form of the frequency per km^2 at $D = 1$ km were calculated automatically.

Measurement areas and craters measured (sizes and location) were calculated and displayed graphically for comparison with pictures, thereby allowing flexibility for adding or subtracting areas.

Data Display

Mainly following suggestions of the report of the *Crater Analysis Techniques Working Group* [1979], we present crater

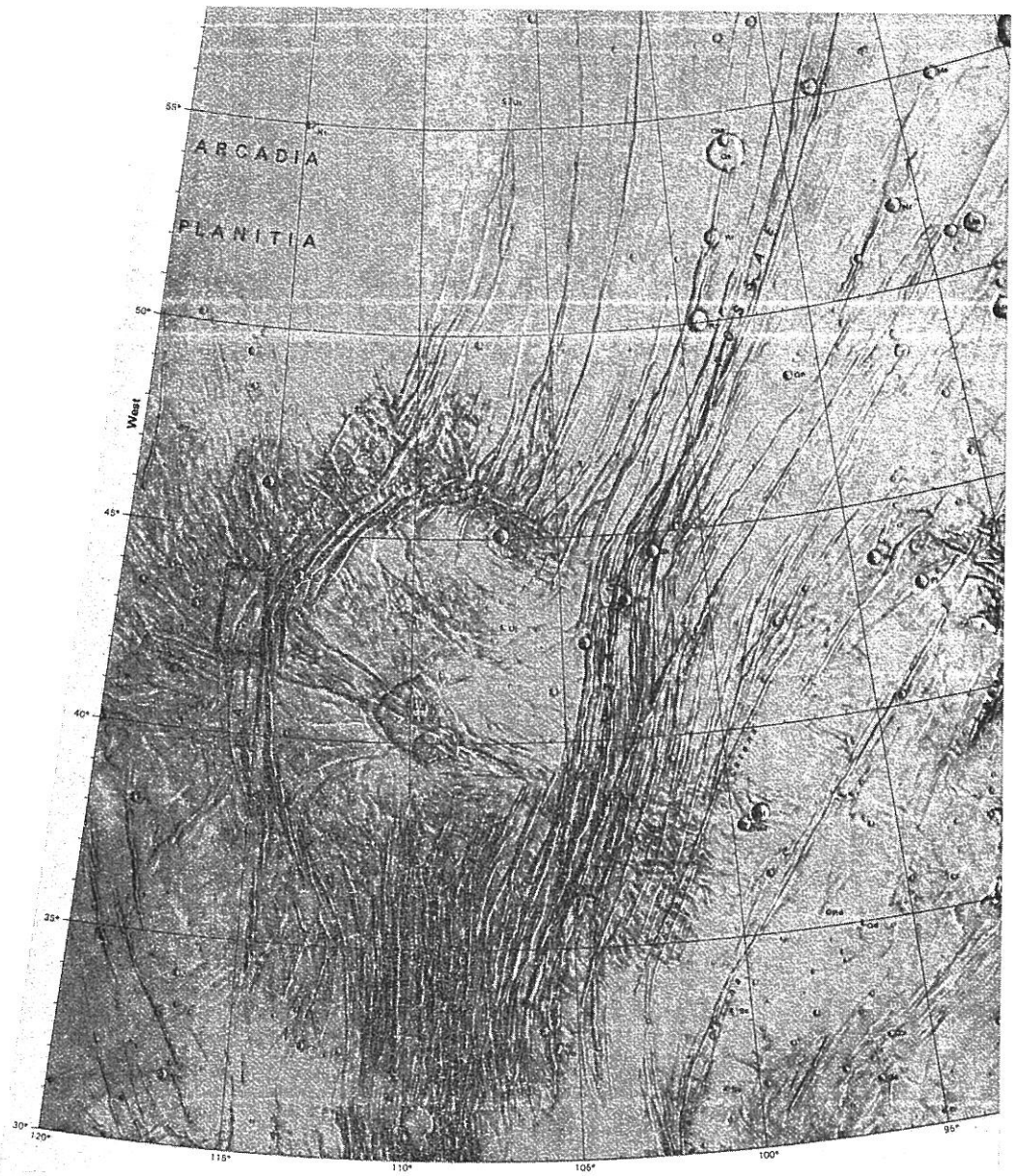


Fig. 12*g*. Alba Patera with areas selected for crater counts: topomap shows selected area for measurements (Figure 11*g*) in western fault zone.

statistics data as cumulative size-frequency distributions and in some cases as differential size-frequency distributions. The cumulative crater size-frequency is defined as the number of craters of a certain size and of all larger craters per unit area, here termed $N(D)$. The differential size-frequency is $dN(D)/dD$, or the approximation $\Delta N/\Delta D$, because we deal with discrete sizes.

The data are presented in quasi-logarithmic increments with respect to crater diameter. There are 18 steps per diameter decade, i.e., in the interval $1 \leq D \leq 10$ we have the steps 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.7, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0. The larger $2^{1/2}$ interval increment suggested by the Crater Analysis Techniques Working Group as the maximum increment size is in fact insufficient for good determination of the distributions in the range of $\lesssim$1-km diameter.

We have put emphasis on a complete presentation in cumulative form. This is the form most appropriate for age determinations because of independence of choice of diameter increment size. The differential size-frequency presentation is useful for enhancing irregularities in the distributions as caused, for example, by resurfacing processes destroying craters. We do not present the data in form of relative frequency distributions because those distributions have a limited value only for distribution index studies (slope of distribution on log-log plot), and for our purposes, analysis of the differential distributions of the data for detection of distribution irregularities will do.

For our own data, information on area location is usually given on uncontrolled mosaics, single pictures, or maps together with frame numbers and latitude and longitude (cf. Figures 11 and 12 and Table 2). Complete size-frequency data have not been given in tabular form (though suggested by the *Crater Analysis Techniques Working Group* [1979]) because they would have occupied too much space. The numerical data may be published or made available (e.g., through the Lunar and Planetary Institute, Houston, Texas).

TABLE 2. Crater Retention Ages (Cumulative Crater Frequency Values $N(1)$ at 1-km Crater Diameter) Extracted From Our Data Given in Figure 11, and Relevant Support Data

Figure	Feature	Viking Photo Number	Position	Slant Range, km	Area, km^2	$N(D = 1$ km), km^{-2}
12*a*	Elysium Mons					
	surroundings	541A 9, 13, 18–20	37.4–24.4°N,219.8–222.6°W	6,524	147,000	$2.0 \cdot 10^{-3}$
	aureole	541A 25, 26, 27	32.1–32.2°N,219.0–216.7°W	6,104	89,300	$1.2 \cdot 10^{-3}$; $2.9 \cdot 10^{-3}$
	shield	541A 46	25.9°N,212.3°W	5,763	25,000	$1.8 \cdot 10^{-3}$; $4.8 \cdot 10^{-3}$
12*b*	Hecatus Tholus					
	surroundings	086A 32, 34, 36, 38	33.3°N,213.4–211.0°W	1,503	5,400	$1.3 \cdot 10^{-3}$
	shield	086A 38, 40, 42, 44	33.1–33.2°N,211.0–208.6°W	1,507	5,600	$1.3 \cdot 10^{-3}$; $1.5 \cdot 10^{-2}$
12*c*	Apollinaris Patera	635A 57	9.1°S,186.4°W	10,013	41,800	$3.5 \cdot 10^{-3}$; $8.8 \cdot 10^{-3}$
12*d*	Arsia Mons					
	caldera rim	422A 29, 38, 39	8.1–9.7°S,121.3–118.9°W	1,541	4,200	$8.6 \cdot 10^{-4}$
	caldera floor	422A 31, 33, 35–37	8.4–9.3°S,120.9–119.4°W	1,555	3,900	$7.1 \cdot 10^{-5}$
12*e*	Ascraeus Mons					
	main caldera floor	401B 19, 21	9.3–9.9°N,104.0–103.8°W	883	530	$1.2 \cdot 10^{-4}$
	small caldera floor	401B 22	10.0°N,103.8°W	883	30	$2.6 \cdot 10^{-4}$
	shield	401B 16, 18, 23	7.9–10.6°N,103.7–104.1°W	893	1,140	$2.9 \cdot 10^{-4}$
12*f*	Ceraunius Tholus	662A 56, 58	24.6–24.2°N,96.3–96.2°W	1,410	1,780	$2.0 \cdot 10^{-3}$; $4.3 \cdot 10^{-3}$
12*g*	Alba Patera	444B 01–04	41.9–42.5°N,115.8–115.9°W	385	350	$9.6 \cdot 10^{-4}$
		444B 05	42.9°N,115.9°W	379	240	$4.0 \cdot 10^{-4}$
		444B 07	43.4°N,116.1°W	376	50	$1.2 \cdot 10^{-4}$
		444B 23, 26, 27	43.7–44.1°N,115.9–115.8°N	404	140	$1.9 \cdot 10^{-3}$

Our crater counts for a variety of volcanic constructs and their surroundings are displayed in Figures 11*a*–11*g*. The respective counting areas are shown in Figures 12*a*–12*g*. Two examples of data for channels and their surroundings are given in Figures 6 and 16. If a unit is covered by more than one Viking image, areas and crater numbers have been added together and marked with one symbol.

Crater retention ages have been extracted from those data in the form of cumulative crater frequency at 1-km crater diameter, $N(1)$, by application of our Martian standard distribution as discussed before. The resulting numerical values are listed in Table 2, together with support data such as size of counting area or location.

Data from other workers were treated as described in the previous sections. In case of obvious numerical errors such as errors in exponential display (e.g., by a factor of 10 in *Carr et al.*'s [1977] data) or the reading of 2-km-diameter values at resolution limits (e.g., Apollinaris Patera [*Plescia and Saunders,* 1979]) giving too low numbers, data have been corrected.

This paper emphasizes the overall history of the planet. Therefore individual volcanoes were not mapped in great detail as has been done, for instance, by *Schaber et al.* [1978]. Only a short discussion of the individual features investigated is given in the following.

Elysium Mons

The uncontrolled mosaic of revolution 541 A covers the western part of Elysium Mons. The caldera is found in the southeastern corner of the mosaic. Three areas have been investigated (Figure 12*a*): Approximately half of the shield material (C) has been covered. The aureole material (B) originally mapped by *Scott and Allingham* [1976] from Mariner 9 images was remapped. The surrounding rolling plains material [*Hiller,* 1979] shows a variety of erosional and permafrost features. Area A is cut by several channels.

Measurements (Figure 11*a*) show that the shield is older (higher crater frequency at all sizes) than the aureole material and the surrounding plains. There is a slight excess of large craters compared with the standard curve. This indicates resurfacing on the shield with survival of the large craters. Therefore two different crater retention ages as in Table 2 have been extracted.

Hecates Tholus

Part of the shield (B) of Hecates Tholus and surrounding plains lavas (A) were measured (Figure 12*b*). The shield and the surroundings are characterized by lava channels with lava flooded impact craters and craters of questionable origin.

Crater counts (Figure 11*b*) clearly show resurfacing, similar to *Plescia and Saunders'* [1979] data (cf. Figure 8). Two different ages are extractable ($N(1) = 1.3 \cdot 10^{-3}$ (km^{-2}) and $N(1) = 1.5 \cdot 10^{-2}$(km^{-2})), indicating that the volcano may have been active over a long period of time. Note that the crater retention age of the survivor craters does not necessarily give the age of the beginning of the resurfacing activity, but it gives the age of emplacement of an older surface. This may have been another phase of activity; that is, there may have been a subsequent period of silence broken only at the time given by the younger ($N(1) = 1.3 \cdot 10^{-3}$ (km^{-2})) age.

Data for the plains (A) follow those of the shield almost exactly; i.e., the plains show crater retention ages very similar to the shield, both for the young and for the old population. Those lava flows therefore are most probably directly connected to the main volcano. More crater counts and detailed mapping of the surrounding plains are needed to resolve this problem.

Apollinaris Patera

The complete shield of Apollinaris Patera was measured including the caldera (Figure 12*c*). Prominent is a broad lava fan to the south that apparently has drained a lava lake through a cut in the caldera rim. Crater counts (Figure 11*c*) show a very flat slope (≈ -1.4) of the curve. This indicates more continuous resurfacing. This means the emplacement of a series of lava flows covering each other and building up the broad fan as it appears now.

Arsia Mons

On Arsia Mons the caldera floor (B) and part of the fractured rim area (A) were measured (Figure 12*d*). The data in

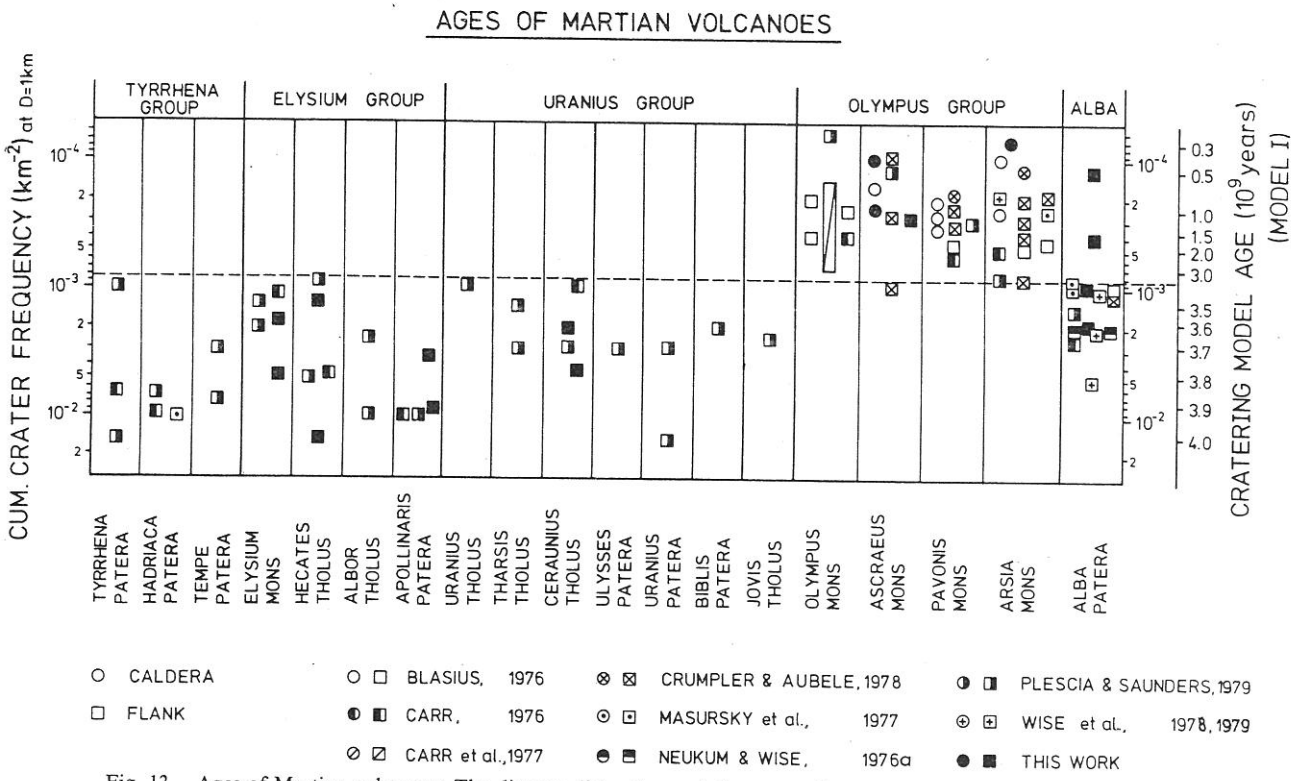


Fig. 13. Ages of Martian volcanoes. The diagram lists all cumulative crater frequency data for 1-km crater diameter (crater retention ages $N(1)$) published by various authors and reinterpreted by us and of own measurements for flanks and calderas of volcanic constructs. $N(1) = 8 \cdot 10^{-4}$ (km^{-2}) subdivides the volcanic activities into an early phase A and a younger phase B.

Figure 11*d* show that the caldera filling is clearly younger than the rim material. The steep slope of the data for the caldera floor indicates that despite careful selection, volcanic craters may have been included in the measurements or that there has been a tendency to enhance diameters at small sizes near the resolution limit. At crater diameters close to the resolution limits these effects can substantially affect the value of crater counts. In our case, only craters with diameters >300 m were used to extract a retention age for the caldera floor.

Ascraeus Mons

Three areas on the summit of Ascraeus Mons were measured (Figure 12*e*): the floor of the central caldera (A), part of the floor of the small side caldera (B), and a section of the rim (C). Data (Figure 11*e*) are consistent with photogeological evidence. Part of the small caldera has collapsed to the lower level of the main caldera. The lavas that make up the floor have been superposed on the collapsed material and therefore are younger than the floor of the small caldera. The flank material is the oldest.

Ceraunius Tholus

Part of the shield of Ceraunius Tholus was measured (Figure 12*f*). The area is dominated by a pattern of lava flows. From the data (Figure 11*f*), two different relative ages have been extracted. Below 1-km crater diameter, resurfacing has extinguished craters.

Alba Patera

Counts were performed for the central part of Alba Fossae (Figure 12*g*). The high resolution allowed precise measurement of craters with diameters as small as 80 m (Figure 11*g*). Two main periods of activity show up in the data. The counts for A and D give the ages of an older activity that made up the bulk of lavas on Alba Patera (cf. Figure 13). Measurements B and C have been made on well-defined single lava flows. The flows are substantially younger than the bulk of Alba Patera and may represent a late stage of volcanic activity.

Ages of Martian Volcanoes

Our data show that the Martian volcanoes had a complicated multistage history with many resurfacing processes that show up in the crater distributions. The different crater retention ages give the different time marks of major events and a relative chronology. Cratering model ages can be inferred from the crater retention ages by application of a model impact chronology.

In the following we shall discuss the relative chronology of Martian volcanoes (as we understand it now) and give estimates of cratering model ages based on our model chronologies.

For detection and understanding of probable global or regional time dependence of volcanism in general or of a specific type, a classification system of all volcanic features is needed. *Plescia and Saunders* [1979], who have carried out crater counts on a variety of volcanoes, have used a system that was established originally by *Pike* [1978] and is based on petrological and morphological criteria.

Three groups are discerned by Plescia and Saunders: highland patera, dome and composite, and basaltic shield. Our own classification mode is somewhat different. The basic criterion is the belonging to regional volcanic provinces like Tharsis or Elysium province or the highlands as a whole. In

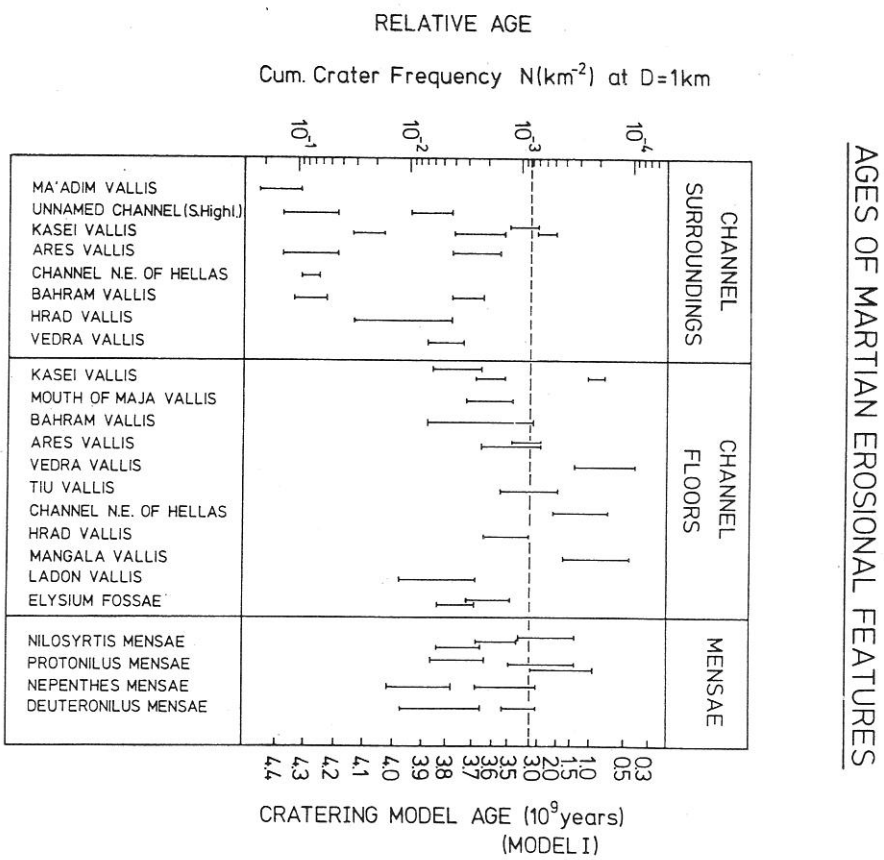


Fig. 14. Ages of Martian erosional features. The diagram lists relative and cratering model ages of mensae channel floors and their surroundings. Data are from *Masursky et al.* [1977], *Malin* [1976], *Squyres* [1978], and *Hiller and Neukum* [1979]. Ages of erosional features are subdivided by $N(1) = 8 \cdot 10^{-4}$ (km^{-2}) into two phases.

this way, time dependence of volcanism in different regions should show up more clearly.

The complexity of Tharsis volcanic province led to a subdivision into two main groups based on the size of the construct (Uranius and Olympus Group). Alba Patera was set aside as a third and separate group because of its unequalled morphology, size, and time span of activity. All groups were named after one typical construct belonging to the specific group. Therefore the Martian volcanic constructs were divided into five groups: Tyrrhena, Elysium, Uranius, Olympus, and Alba.

For the description of the volcanic activity of a group, two terms are used: 'overall activity' means a time span limited by the youngest and the oldest surface found in a group, and 'main activity' describes a core time when most volcanoes of a group were active. The crater retention ages ($N(1)$ values) extracted from our crater frequency data of volcanic shields and caldera floors and the reinterpreted data of other workers are displayed in Figure 13 in accordance with our classification system. The data will be discussed in the following.

Tyrrhena Group

The Tyrrhena Group is named after Tyrrhena Patera and in addition consists of Hadriaca Patera and Tempe Patera. All volcanoes are located in the highlands (hilly and cratered material) and therefore are situated on the oldest crust of the planet. Low profile and intense channeling of the flanks are common attributes. Crater statistics of their flanks show values of $N(1) = 3 \cdot 10^{-3}$ to $1.5 \cdot 10^{-2}$ (km^{-2}) for the overall activity and $N(1) = 6 \cdot 10^{-3}$ to $1 \cdot 10^{-2}$ (km^{-2}) for the main activity, which corresponds to cratering model ages of 3.8–4.0 b.y. on the basis of our model I cratering chronology (cf. Figure 9*a*), which will be applied in the following if not stated otherwise.

The time of activity as documented by the crater counts is consistent with the emplacement of the younger plains lavas (unit cratered plateau material) found in several areas of the southern highlands [*Hiller,* 1978, 1979]. Surfaces with crater numbers of $N(1) = 6 \cdot 10^{-3}$ (km^{-2}) or absolute ages of 3.8 b.y. seem to be the youngest found in the highlands and therefore seem to mark the end of substantial volcanism in the southern hemisphere of the planet.

Elysium Group

The Elysium volcanoes Elysium Mons, Hecates Tholus, and Albor Tholus are located in the northern plains on top of the Elysium bulge of the Martian crust. They show the typical forms of shield volcanoes. Their shields are partially embayed by younger plains lavas, and in other cases their lava flows bury the surrounding plains [*Schaber et al.,* 1978]. Apollinaris Patera is located in the northern plains southeast of the Elysium volcanic province, i.e., not directly on the Elysium bulge. We have added it to the Elysium volcanic group because of its proximity and contemporaneity of activity.

Crater counts on the shields of the Elysium volcanoes yield crater retention ages of $N(1) = 9 \cdot 10^{-4}$ to $1 \cdot 10^{-2}$ (km^{-2}) or cratering model ages of 3.2–3.9 b.y. One single count (Hecates Tholus) shows a crater retention age of $N(1) = 1.5 \cdot 10^{-2}$ (km^{-2}), probably representing the underlying old cratered plains unit visible through relatively thin cover of shield material.

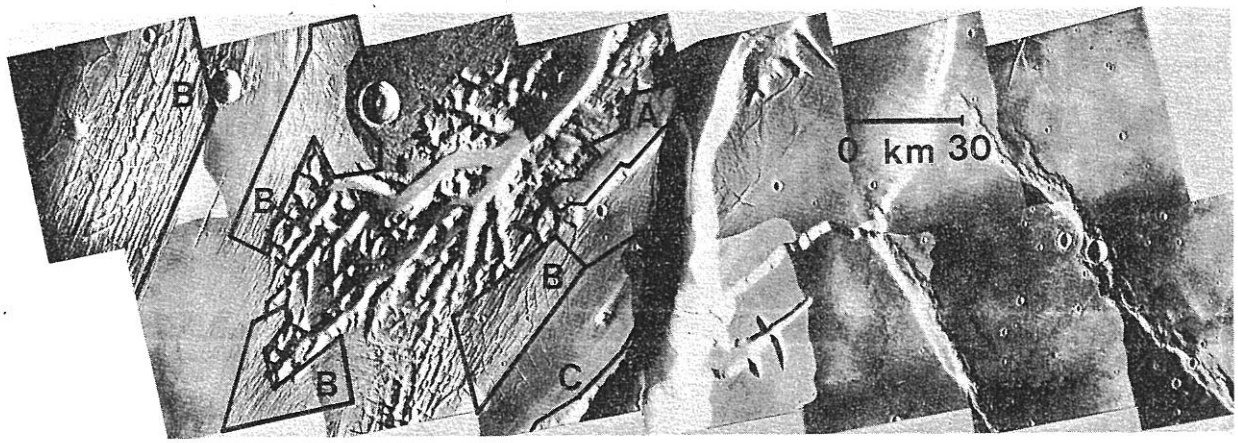


Fig. 15*a*

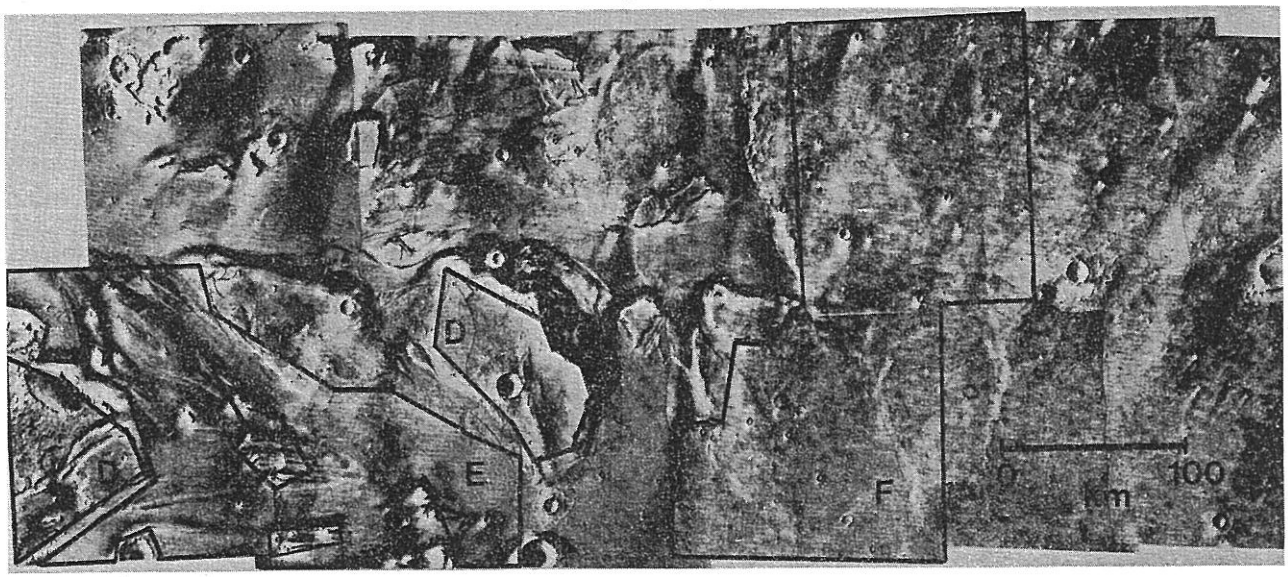


Fig. 15*b*

Fig. 15. Kasei Vallis, measurement areas. Data are given in Figure 16. (*a*) West end (23°N, 71°W) is cut into Lunae Planum. The plains are shaped by some erosional activity. (*b*) Mouth of Kasei Vallis into Chryse Planitia (27°N, 52°W). Stream islands show cratering records of Lunae Planum.

The ages of the volcanic shields are nearly coincident with the ages of the surrounding units smooth plains material and cratered plains material [*Hiller and Neukum,* 1979]. Emplacement of volcanic constructs and plains lavas at the same time interfering with each other means the production of magmas of at least two types of chemical composition or an alternation of high and low eruption rates of the volcanoes [*Womer and Greeley,* 1980]. The volcanic activity of the Elysium region also has to be seen as part of the emplacement of lavas on a global scale during a time span marked by crater retention ages of $N(1) \approx 1 \cdot 10^{-3}$ to $1 \cdot 10^{-2}$ (km^{-2}).

Tharsis Volcanic Province

The Tharsis region is the most prominent volcanic province on Mars. Following our criterion of local dependence, all volcanic constructs of that area should be represented by one single group. Because of the high number and wide type spectrum of those volcanoes and in order to gain more flexibility for age comparisons, the following subdivisions were made.

Uranius group. The Uranius group covers the small constructs (80- to 200-km diameter) of Tharsis volcanic province. It was named after Uranius Patera and Uranius Tholus and after *Plescia and Saunders* [1979] represents two different types of volcanism: first, shield volcanoes formed by multiple flows of basaltic lavas with gentle slopes (basaltic shield group)—Uranius Patera, Biblis Patera, Jovis Tholus—and second, constructs built by lavas more viscous than basalt (dome and composite group)—Uranius Tholus, Tharsis Tholus, Ceraunius Tholus, and Ulysses Patera. Crater retention ages of the first subgroup Uranius Patera, Biblis Patera, and Jovis Tholus lie between $N(1) = 3 \cdot 10^{-3}$ and $2 \cdot 10^{-3}$ (km^{-2}), equivalent to ages of 3.7 to 3.6 b.y. A data point for Uranius Patera of $N(1) = 1.5 \cdot 10^{-2}$ (km^{-2}) is considered to be representative of a crater retention age of the prevolcanic surface unit.

Crater data of the second subgroup range from $N(1) = 4.5 \cdot 10^{-3}$ (km^{-2}) to $1 \cdot 10^{-3}$ (km^{-2}), which equals 3.8 to 3.3 b.y., the main activity of that group lying between $N(1) = 3 \cdot 10^{-3}$ (km^{-2}) and $N(1) = 2 \cdot 10^{-3}$ (km^{-2}), or 3.7 to 3.6 b.y. ago.

Dating of the volcanoes of the Uranius group shows that the period of activity of the individual constructs was relatively short, with the exception of Ceraunius Tholus showing numbers ranging from $N(1) = 4.5 \cdot 10^{-3}$ to $1 \cdot 10^{-3}$ (km^{-2}), or 3.8 to 3.3 b.y.

Olympus group. The big basaltic shield volcanoes Olympus Mons, Ascraeus Mons, Pavonis Mons, and Arsia Mons have been assembled in the Olympus group, named after Olympus Mons, the most prominent volcanic feature on Mars. With heights of up to 25 km and shield diameters between 400 and 600 km; the dimensions of the Olympus volcanic constructs are outstanding in the solar system. Crater counts show numbers of $N(1) = 1 \cdot 10^{-3}$ to $6 \cdot 10^{-5}$ (km^{-2}), which means cratering model ages of 3.3 to 0.3 b.y. Main activity is characterized by numbers of $N(1) = 6 \cdot 10^{-4}$ to $1.3 \cdot 10^{-4}$ (km^{-2}) (2.5–0.5 b.y.). The cratering data reveal relatively great similarity of surface ages of the various volcanic constructs of the group. Especially in this group many more data are needed in order to establish a detailed chronology of the surface units of those volcanoes and of the lava flows extending far out into the surrounding plains [*Schaber et al.,* 1978; *Lopes et al.,* 1980].

Alba group. Alba Patera was set aside as a single 'group' because of its unique size, morphology, and time span of activity. Data shown in Figure 13 are from all parts of the volcano including the old caldera within the fault zone but not from the younger inner caldera. Crater counts range from $N(1) = 5 \cdot 10^{-3}$ to $1.2 \cdot 10^{-4}$ (km^{-2}), or 3.8 to 0.5 b.y. This means activity of this one volcano over a period of at least 3.3 b.y.

The available data also show different phases within the overall activity, the bulk of activity lasting from $N(1) = 5 \cdot 10^{-3}$ to $8 \cdot 10^{-4}$ (km^{-2}), or from 3.8 to 3.2 b.y. ago, and younger eruptions occurring at $N(1) = 4 \cdot 10^{-4}$ and $1.2 \cdot 10^{-4}$ (km^{-2}), or 1.7 and 0.5 b.y. ago. The young activity seems to consist of isolated lava flows. At least in the area measured, the young sequence produced a series of relatively thin and isolated lava flows.

The gaps between the different times of activity may be real and would mean repose periods of the order of a billion years at times <3.5 b.y. ago. Our data indicate that the volcanism of Alba Patera was partially revived after a long period of nonproduction at the time of maximum activity of the Olympus group. It has to be kept in mind, though, that these results are somewhat speculative because of the few data points existing up to now. Alba Patera is a huge area, and a great number of crater counts on small homogeneous surfaces and on individual lava flows have still to be performed to cover the history of this volcanic construct in detail.

Comparing all data compiled in Figure 13, a crater retention age of $N(1) \approx 8 \cdot 10^{-4}$ (km^{-2}) seems to subdivide all constructs into an early (phase A) and a late (phase B) volcanism. The only exception is Alba Patera, whose activity overlaps that of both epochs.

Phase A volcanism built up the Tyrrhena group including all highland volcanoes, the Elysium group consisting of the volcanoes of the Elysium region including Apollinaris Patera, and the Uranius group with all small constructs of the Tharsis area. Phase B is confined to the volcanism that built up the shield volcanoes of the Olympus group. Alba Patera seems to show an early period of activity that is part of phase A and would be connected to the old Tharsis volcanoes of the Uranius group. After a period of nonactivity a revival of lava production follows, connected to phase B with the piling up of the young Olympus volcanoes.

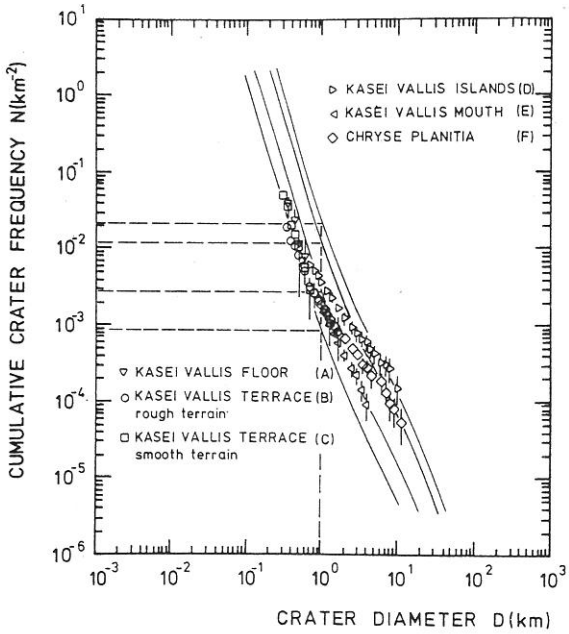


Fig. 16. Kasei Vallis, crater size-frequency data (areas of Figure 15). Four different ages of units can be interpreted by fitting the standard curve.

Ages of Erosional Features

Crater frequencies have been measured for Martian channels (floors and surroundings) and for some mensae at the boundary between Southern Highlands and Northern Plains. We shall not discuss occurrence and morphology of Martian channels. The reader is referred to the literature [e.g., *Masursky et al.,* 1977]. The term 'mensae' is used to describe the highland/lowland boundary area west of 180° through 30° longitude, because the term 'knobby material' [*Scott and Carr,* 1978] is not limited to a certain area. Examples of our counts are given in Figures 6 and 16. Data from *Masursky et al.* [1977], *Malin* [1976], and *Squyres* [1978] have been used and reinterpreted as described before, and the results together with our own data are given in Figure 14.

Interpretation is as follows:

1. Areas surrounding the highland channels show two periods of surface emplacement, the older one ($N(1) > 4 \cdot 10^{-2}$ (km^{-2}) or > 4.2 b.y.) representing the early crust of the planet. The younger ages mark a resurfacing period with crater retention ages of $N(1) \approx 2 \cdot 10^{-3}$ to $2 \cdot 10^{-2}$ (km^{-2}) or 3.6 to 4.1 b.y.
2. Channel floors show a variety of formation ages ranging from $N(1) = 1 \cdot 10^{-2}$ (km^{-2}) to $1 \cdot 10^{-4}$ (km^{-2}) or from 3.9 to 0.4 b.y., the bulk lying between $N(1) = 5 \cdot 10^{-3}$ (km^{-2}) and $8 \cdot 10^{-4}$ (km^{-2}), or 3.8 and 3.1 b.y.

Consistent with the history of volcanic constructs, there is evidence that channel resurfacing was going on during two separate phases. Again $N(1) \approx 8 \cdot 10^{-4}$ (km^{-2}), or a cratering model age of 3.1 b.y., seems to mark the end of the first phase (phase A). At least in the case of the north fork of Kasei Vallis (see Figure 7) it can be shown that only the older crater retention ages mark the actual channel excavation process. The young surface of the channel floor with resurfacing at $N(1) = 8 \cdot 10^{-5}$ (km^{-2}) was produced by a late filling, which most probably consisted of low-viscosity lavas produced at the time

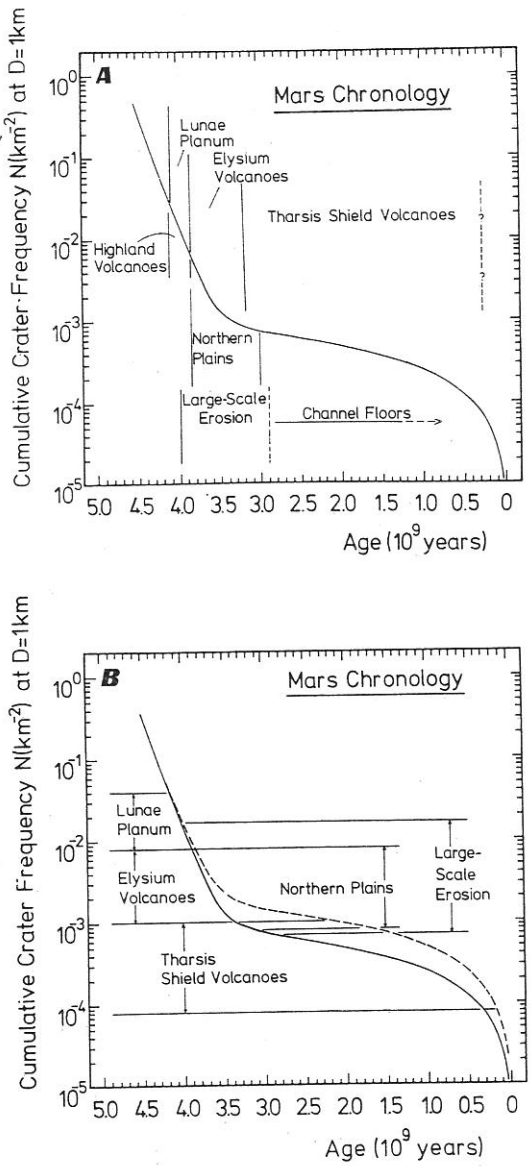


Fig. 17. Mars chronology. The diagrams present an overview of Martian history as derived from measured and reassessed crater size-frequency data. (*a*) Data based on model I Mars cratering chronology (Figure 9*a*); boundaries between different epochs are only approximate and in most cases activities overlap. (*b*) Model I and model II cratering chronologies compared. The diagram shows the consequences for absolute ages of major events on Mars if the different cratering chronologies as in Figure 9*b* are used (model I versus a model II cratering chronology).

of phase B volcanism (Tharsis shield volcanoes) (also revival of Alba Patera during that same time span). Kasei Vallis, south fork (Figure 15), gives an example for the extraction of age data from the crater size-frequency distributions for different episodes and magnitude of resurfacing activity in a channel system. Measurements have been performed for the channel floor, for the adjacent channel terraces, for stream islands, for the mouth of the channel into Chryse Planitia, and for some of the adjacent plains area. The cumulative size-frequency distributions are displayed in Figure 16. The distributions do not follow the standard distribution (solid curves) but show various degrees of flattening toward small sizes.

The oldest age in the Kasei Vallis system is marked by the surface of the stream islands (Figure 15, areas D). The large-size craters with retention ages $N(1) \approx 2 \cdot 10^{-2}$ (km^{-2}), or 4.1 b.y. (Figure 16), are survivors from the original highland plains surface (Lunae Planum). The magnitude of the resurfacing activity is expressed in the numbers of survivor craters, i.e., in the different frequency of large craters in the different areas measured. The data in Figure 16 show that more large craters survived on the islands of Kasei Vallis than in any of the other areas. This means that the islands were least affected by the resurfacing activity. This is supported also by the higher number of craters around 1-km diameter. The other areas measured show various degrees of resurfacing according to their different large-size crater retention ages. All data, however, tend to approach asymptotically the same crater frequency curve at small sizes ($D < 1$ km) approximately given by the lowest standard curve in Figure 16 with a crater retention age of $N(1) = 9 \cdot 10^{-4}$ (km^{-2}). The intermediate curve that is fitted to the data for Chryse Planitia (F) shows an age of $N(1) = 1.2 \cdot 10^{-2}$ (km^{-2}) for crater diameters larger than 6 km; the one fitted through the data for the mouth of Kasei Vallis (E) gives $N(1) = 3 \cdot 10^{-3}$ (km^{-2}). The mouth of the channel therefore is younger at large crater sizes than the adjacent plains materials of Chryse Planitia, which should be expected because features of erosion traced out into Chryse Planitia become fainter farther away.

3. The mensae region (Deuteronilus, Protonilus, Nilosyrtis, Nepenthes mensae) is made up of knobby and chaotic terrain over a length of 10,000 km [*Carr et al.*, 1973]. The highland scarp has been eroded backward, leaving behind rectangular mesas or clusters of zeugenberge.

The time of occurrence of this erosional activity can be determined only by indirect evidence. The surface age of the eroded highland scarp must be preerosional, and that of the eroded areas (in most cases flooded plains lavas) post erosional. *Hiller* [1978] has shown that adjacent plains lavas which extend into the eroded area are as old as the young highland lavas. The maximum period of retreat of the highland scarp therefore is given by the crater retention ages of the cratered plateau material and of the cratered plains material. Both units show crater retention ages of $N(1) = 4 \cdot 10^{-3}$ to $1 \cdot 10^{-2}$ (km^{-2}), or 3.7 to 4.0 b.y., and therefore are of the same age within the accuracy of the data. This means that the erosional process took place in a relatively short period of time, probably between 100 and 300 m.y. After this major period of erosion only minor erosional activity and further flooding of the plains with low-viscosity lavas (smooth plains material) took place ($N(1) = 8 \cdot 10^{-4}$ to $3 \cdot 10^{-3}$ (km^{-2}), or 3.1–3.7 b.y.).

Younger ages of $N(1) = 8 \cdot 10^{-4}$ to $3 \cdot 10^{-4}$ (km^{-2}) measured in the eroded area may be due either to permafrost processes [*Carr and Schaber*, 1977; *Hiller*, 1978] or to continuation of lava production on a smaller scale. Because of the evidence of permafrost features and the apparent lack of lava source regions, a resurfacing by permafrost processes is considered to be more likely.

CONCLUSIONS

The age data of Martian volcanic and erosional features, which we have treated and compiled here, may be used to obtain an impression of the bulk history of the planet Mars. An overview of the time relations of major geologic events on Mars on the basis of our data is given in Figures 17*a* and 17*b*. The chronology of Martian geological events as given in Figure 17*a* is based on our model I cratering chronology of Figure 9*a*. The age limits for individual events have been given in

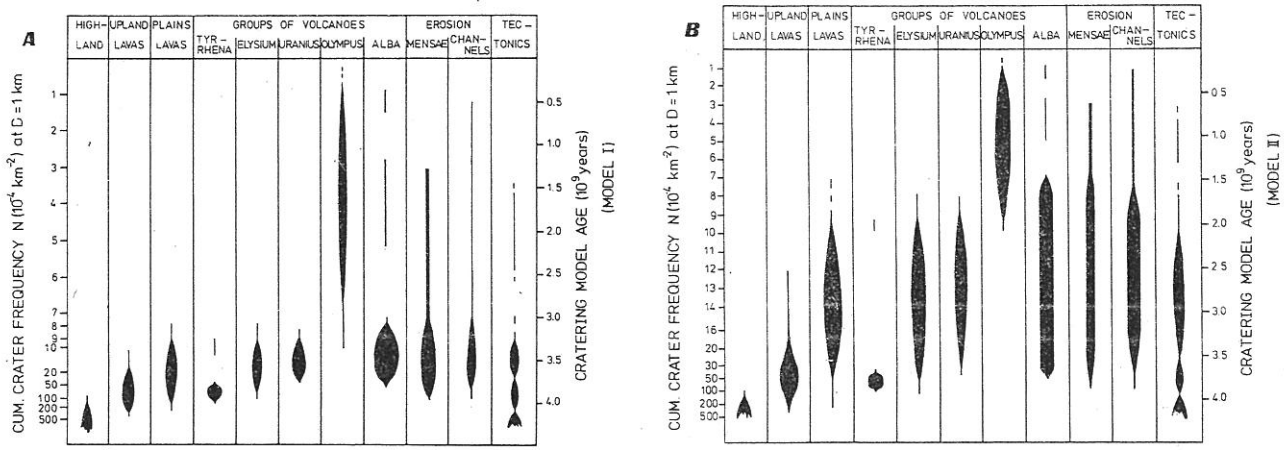


Fig. 18. Martian geologic activity. The diagrams show in a qualitative way cratering model ages and periods of geologic activity. (*a*) Cratering model ages based on model I cratering chronology (cf. Figures 9*a* and 17*a*). Notice that most major events took place before 3.0 b.y. ago. Exception is the Olympus volcanoes group and Alba Patera, active through the recent history. Channel ages of approximately 1 b.y. mark young floor filling but not original excavation. (*b*) Cratering model based on model II cratering chronology (cf. Figures 9*b* and 17*b*). This chronology gives longer periods of activity of the individual geological event. As in Figure 18*a*, separation into two epochs of activity (phase A and B) is still detectable (division here at an age of about 1.5 b.y.).

cratering model ages there to emphasize absolute relationships. The age limits of the individual events are only approximate, and the common large overlaps in time are not shown. Figure 17*b* gives an idea of the dependence of the data on the cratering chronology used (model I or model II).

A better view of the geological history of Mars and of its periods of activity can be obtained if data are presented as in Figure 18*a*. The times of occurrence of major geological events are given in a very qualitative way for highland and plains lavas, constructional volcanic materials, erosional features, and tectonism. For better comparison, data are not displayed in logarithmic format (suited best for presenting ages in form of crater frequency data) but on a linear scale to emphasize absolute age relationships. This compilation has to be taken as a first attempt at comparing all crater retention ages available and tractable by our methodology. It may have to be revised according to new data and because of its dependence on possible adjustments of the model chronologies. Figure 18*a* suggests that the planet was highly active during its early history. During this time span (before 3.0 b.y. ago) all important events like highland resurfacing and volcanism, emplacement of the plains, erosion of the highland scarp and of channel systems, and most of the tectonic events in the Tharsis region occurred. After this period of high activity the development of the planet was largely limited to the volcanism of the Tharsis bulge.

The simultaneous occurrence of intense lava production and large-scale erosion (probably caused by melting of subsurface ice layers [*Carr and Schaber,* 1977] on a global scale supports models like the one published by *Wise et al.* [1979]: Global heating in the early history leads to intense production of low-viscosity lavas, to tectonic events caused by mantle/crust processes, and to the melting of permafrost ice and subsequent erosion of the resulting debris. After cooling down, global activity is replaced by minor regional activities active until relatively recently. Crater retention ages of $N(1) \approx 8 \cdot 10^{-4}$ (km^{-2}) or, on the basis of the model I cratering chronology, the age of 3.0 b.y. seems to mark the end of the older phase of Martian history (phase A) and the beginning of younger Martian history (phase B).

In conclusion, the data can be interpreted as follows:

1. Volcanic activity on Mars lasted over an extremely long period of time, probably until 0.3 b.y. ago or less.

2. Erosional activity peaked with the retreat of the highland scarp ($N(1) = 1 \cdot 10^{-2}$ to $4 \cdot 10^{-3}$ (km^{-2}), or 4.0 to 3.7 b.y. ago). Channel formation took place between $N(1) = 6 \cdot 10^{-2}$ to $7 \cdot 10^{-4}$ (km^{-2}), or 3.8–3.1 b.y. ago. Young ages measured on channel floors most probably are posterosional fillings.

3. Three major periods are distinguishable: first period (ages $N(1) = 1.5 \cdot 10^{-2}$ to $3 \cdot 10^{-3}$ (km^{-2}), or 4.0 to 3.7 b.y.): final resurfacing of the highlands (cratered plateau material), shield volcanoes in the highlands (Tyrrhena group), intense erosional retreat of the highland/lowland boundary areas (knobby/chaotic material), emplacement of the old plains lavas (cratered plains material), erosion of some channels; second period (ages $N(1) = 3 \cdot 10^{-3}$ to $8 \cdot 10^{-4}$ (km^{-2}), or 3.7 to 3.0 b.y.): Elysium volcanoes, main activity of Alba Patera, small Tharsis volcanoes (Uranius group), emplacement of younger plains lavas (smooth plains material and rolling plains material), thinning out of erosional activity in the highland/lowland boundary, bulk of channel formation; third period (ages $N(1) = 8 \cdot 10^{-4}$ to $6 \cdot 10^{-5}$ (km^{-2}), or 3.0 to 0.3 b.y.): piling up of the big Tharsis volcanoes, Alba Patera revival, channel fillings.

This picture is consistent also with models of the evolution of the Martian atmosphere. A rapid decrease of atmospheric pressure 1 b.y. ago [*McElroy et al.,* 1977] necessarily limits liquid surface water and substantial eolian erosional activities to the time before.

Absolute time of geological events based on cratering data still is a controversial issue. All absolute ages given so far used a crater chronology model originally established by *Neukum and Wise* [1976*a*] and were applied here in a somewhat modified version (model I). As discussed in the first part of this paper, an uncertainty of a factor of 2–3 in crater frequency in our model chronologies is still possible. (The extracted crater retention ages themselves, however, have uncertainties of usually around ±30%.) This is illustrated in Figure 17*b*, where the application of the model I cratering chronology of Figure 9*a* is compared with the application of the model II cratering

chronology of Figure 9*b* (factor of 2 higher Martian cratering rate for recent times). At the old end, both curves are very close. A curve shift therefore mainly affects the timing of events for crater retention ages of $N(1) \lesssim 2 \cdot 10^{-3}$ (km^{-2}), or 3.5 b.y. Within this time span most of the major geologic activities on Mars occurred.

The consequences for the geologic history of Mars of the application of the model II impact chronology are illustrated in Figures 17*b* and 18*b*: (1) Piling up of the Elysium volcanoes would not be terminated 3.4 b.y. but rather 2.2 b.y. ago. (2) Emplacement of the northern plains would last until 1.6 b.y. ago instead of 3.1 b.y. (3) Large-scale erosion like scarp retreat and channeling would be active from 4.0 b.y. ago until 1.5 b.y. instead of until 3.0 b.y. ago. (4) The development of the big Tharsis shield volcanoes (Olympus group) would be shifted to more recent times, from 3.4–0.3 b.y. to 2.2–0.15 b.y. ago.

A Martian chronology based on this model II cratering chronology would stretch the time span of major geological activities and make them somewhat younger (a factor of 2 for ages <3 b.y.). This would mean a more balanced global geology in a 'traditional' sense, better corresponding with thermal history models such as given by *Toksöz and Hsui* [1978] and *Arvidson* [1980] and would still be in agreement with current models of Martian atmospheric history [e.g., *Arvidson,* 1980].

Even though we still assume that equal lunar and Martian impact mass fluxes are more probable and have preferred our model I cratering chronology for inference of cratering model ages, the truth may be somewhere in between those two model chronologies or somewhat beyond. But regardless of this the major conclusions of this paper remain valid. Most geologic processes are restricted to the earlier times of the planet, very probably before 1.5 b.y. ago, with the exception of the Tharsis shield volcanoes including Alba Patera and some channel fillings. This means that Mars is not a youthful planet but an ancient one with respect to most of its surface features, as already pointed out by *Neukum and Wise* [1976*a*, *b*]. Application of our model II cratering chronology sets the age of the youngest surfaces there on less than 200 m.y. Since we have not yet been able to cover the surface of Mars, other even more recent activity may have remained undetected.

More crater statistics age data, a better established standard crater curve for small diameters, and most important, a better knowledge of the recent cratering rates at Mars are needed to understand its recent history better.

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